

Rice stripe virus utilizes an *Laodelphax striatellus* salivary carbonic anhydrase to facilitate plant infection by direct molecular interaction

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eLife Assessment

This study presents a **useful** set of experiments to explore how a salivary protein might facilitate planthopper-transmitted rice stripe virus infection by interfering with callose deposition and if fully validated, these findings would significantly advance our understanding of tripartite virus-vector-plant interactions and could be of broad interest to plant science research. The authors provide additional data supporting protein-protein interactions and clarify the transient presence of LssaCA in plants. However, the mechanistic framework remains **incomplete**, particularly regarding the temporal dynamics of callose function and the sustained effect of LssaCA after virus inoculation. Evidence for the tripartite interaction's functional relevance is still limited, and several critical phenotypic and biochemical details require further substantiation.

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Abstract

Plant viruses typically rely on insect vectors for transmission between plants, with insect salivary proteins playing critical roles in this process. In this study, we demonstrate how *Laodelphax striatellus* salivary carbonic anhydrase (LssaCA) promotes rice stripe virus (RSV) infection in plants. We discovered that LssaCA directly binds to RSV nucleocapsid protein (NP) in insect salivary glands. This LssaCA-NP complex interacts with a rice thaumatin-like protein (OsTLP) that possess endo- β -1,3-glucanase activity potentially degrading callose. Upon binding, the LssaCA-NP complex significantly enhances OsTLP enzymatic activity. We further clarify that both *L. striatellus* feeding and RSV infection induce callose deposition. The tripartite LssaCA-NP-OsTLP interaction enhances callose degradation, thereby facilitating RSV infection via its insect vector. This study provides new insights into complex virus-insect-plant tripartite interactions mediated by insect salivary proteins, with broad implications for numerous plant viruses transmitted by insect vectors.

Introduction

During plant virus transmission by piercing-sucking insects, most viruses are inoculated into the plant phloem via the insect's secreted saliva (Arca & Ribeiro, 2018 [↗](#); Conway et al., 2016 [↗](#); Wu et al., 2022 [↗](#)). Thus, insect saliva acts as an interface for the virus-insect-host tripartite interaction and can directly promote viral transmission to, and infection of, the host plants (Sun et al., 2020 [↗](#); Wu et al., 2022 [↗](#)). However, despite the importance of insect salivary proteins in this tripartite interaction, there is still much to learn about how these proteins enable successful viral infection.

Previous studies have revealed that there are two ways in which insect saliva facilitates viral infection. One is an indirect approach whereby the saliva modulates the host microenvironment at the feeding site, and saliva effectors work together to allow the arthropod to go unnoticed while it feeds on the host plant (Acevedo et al., 2019 [↗](#); Arca & Ribeiro, 2018 [↗](#); Sun et al., 2020 [↗](#)). For example, our recent work indicated that a *Laodelphax striatellus* mucin protein, LssaMP, enables the formation of the salivary sheath and facilitates the transmission of rice stripe virus (RSV) into the rice phloem (Huo et al., 2022 [↗](#)). The study on leafhoppers revealed that the expression of a saliva calcium-binding protein is inhibited by rice gall dwarf virus (RGDV), thus causing an increase of cytosolic Ca^{2+} levels in rice and triggering callose deposition and H_2O_2 production. This increases the frequency of insect probing, thereby enhancing viral horizontal transmission into the rice phloem (Wu et al., 2022 [↗](#)). The other mechanism by insect saliva to facilitate virus infection is direct regulation, whereby saliva proteins promote virus transmission through specific molecular interactions (Wen et al., 2019 [↗](#)). Direct saliva protein-pathogen interactions have been reported in animal pathogens. For example, during transmission of *Borrelia burgdorferi* by *Ixodes scapularis*, the saliva protein Salp15 of *I. scapularis* binds to the bacterial outer surface protein C, which prevents the bacterium from being recognized by the animal immune system. In this way, the saliva protein enables the pathogen to infect the animal host (Ramamoorthi et al., 2005 [↗](#); Schuijt et al., 2008 [↗](#)). Although most plant viruses are heavily dependent on insect vectors for plant-to-plant transmission (Gray, 2008), the direct function of insect saliva proteins in mediating virus transmission remains largely uninvestigated.

During sap-feeding, arthropods produce two distinct types of saliva at different stages of the feeding process: gel saliva and watery saliva (Bonaventure, 2012 [↗](#); Lou et al., 2019 [↗](#)). The former forms a salivary sheath to provide a smooth path for the stylet penetration (Lou et al., 2019 [↗](#)). The latter is mainly secreted into the phloem sieve elements to prevent them from plugging up and suppress plant defense responses. Some salivary components may act as herbivore-associated molecular patterns that can trigger pattern-triggered immunity, and certain salivary effectors may be recognized by plant resistance proteins to induce effector-triggered immunity, etc. (Huang et al., 2017 [↗](#); Ji et al., 2017 [↗](#); Yi et al., 2021 [↗](#)).

For phloem-feeding insects, callose deposited on phloem sieve plate and plasmodesmata of sieve elements functions as a defense mechanism by reducing insect feeding and preventing viral movement (Hao et al., 2008 [↗](#); Hipper et al., 2013 [↗](#); Will & Vilcinskis, 2015 [↗](#); Zavaliev et al., 2011 [↗](#); Yue et al., 2022 [↗](#)). Callose is a β -(1,3)-D-glucan polysaccharide that is synthesized by callose synthases and degraded by β -(1,3)-glucanases. Plants defend themselves by depositing callose at the sieve plates and plasmodesmata in response to virus infection, whereas viruses counter this defense by activating β -(1,3)-glucanases to degrade callose (Bucher et al., 2001 [↗](#); Hao et al., 2008 [↗](#); Wu et al., 2022 [↗](#); Zavaliev et al., 2011 [↗](#)).

RSV is the causative agent of rice stripe disease, a serious disease of rice crops that has occurred repeatedly in China, Japan, and Korea (Xu et al., 2021 [↗](#)). RSV is completely dependent on insect vectors for transmission among its host plants, and *L. striatellus* is the main vector (Xu et al., 2021 [↗](#); Zhao et al., 2018). *L. striatellus* transmits RSV in a persistent-propagative manner. The

virus initially infects the midgut, then disperses from the hemolymph into the salivary glands, and is inoculated into the plant host during *L. striatellus* feeding (Huo et al., 2022 [↗](#)). *L. striatellus* belongs to the order Hemiptera, whose members mainly feed from sieve tubes through their mouthparts (stylets) that penetrate plant tissues and reach sieve tubes to ingest the phloem sap (Tjallingii, 2006 [↗](#); Van Bel & Will, 2016 [↗](#)). RSV is mainly secreted into the rice phloem via the watery saliva (Huo et al., 2022 [↗](#); Wang & Blanc, 2021 [↗](#)).

In this study, we identified a molecular interaction among RSV, an *L. striatellus* saliva protein, and a plant β -1,3-glucanase. The insect saliva protein directly binds to the RSV nucleocapsid protein (NP) and then binds to a rice thaumatin-like protein to activate its β -1,3-glucanase activity. The activation of β -1,3-glucanase helps RSV infection by inhibiting callose deposition in response to viral infection.

Results

L. striatellus salivary carbonic anhydrase directly bind to RSV NP

Insect-secreted watery saliva is inoculated into plant phloem sieve tubes along with RSV (Huo et al., 2022 [↗](#)). We identified the salivary proteins that directly interact with RSV to promote viral plant infection. Among the saliva proteins identified in our previous study from RSV-free *L. striatellus* (Huo et al., 2022 [↗](#)), Sap6 (GenBank accession No. RZF48846.1) was found to co-localize with RSV in infected salivary glands, suggesting a potential molecular interaction between them (Fig. 1A [↗](#) and S2 [↗](#)).

Sequence homology analysis revealed Sap6 as a carbonic anhydrase-like protein, with a signal peptide located at the N-terminus (amino acids 1-24) (Fig. 1B [↗](#) and S1A [↗](#)). We recombinantly expressed this protein in *E.coli* or in sf9 cells as a secreted protein. The purified proteins were measured enzymatic activity in hydrolyzing *p*-nitroethyl acetate. The observed hydrolyzing activity confirmed Sap6 as a carbonic anhydrase (abbreviated as CA, Fig. 1C [↗](#)). Sequence analysis indicated seven residues constituting the enzymatic active center: H₁₁₁, N₁₃₉, H₁₄₁, H₁₄₃, E₁₅₃, H₁₆₆, and T₂₅₃. We then constructed a mutant with substations of all seven residues (Fig. S1B. [↗](#) H111D, N139H, H141D, H143D, E153H, H166D, and T253E). The mutant protein lost enzymatic activity, indicating the necessity of the enzymatic active center (Fig. 1C [↗](#)). Sap6 was subsequently named *L. striatellus* salivary carbonic anhydrase (LssaCA).

We performed pull-down and Microscale thermophoresis (MST) assays to confirm the molecular interaction between LssaCA and RSV NP. Pull-down assays revealed a specific interaction between LssaCA and NP (Fig. 1D [↗](#)). MST assays determined that the K_D of the interaction between LssaCA and NP was 2.7 ± 2.2 μ M, further confirming the molecular interaction (Fig. 1E [↗](#) and S3 [↗](#)).

We then performed quantitative reverse-transcriptional PCR (RT-qPCR) to determine the gene expression and western blotting to assess protein distribution. RT-qPCR revealed specific expression of *LssaCA* in *L. striatellus* salivary glands (Fig. 1F [↗](#)), while western blotting confirmed the protein's presence in this tissue (Fig. 1F [↗](#)).

When *LssaCA* expression was knocked down using *LssaCA*-specific double-stranded RNA (dsLssaCA) microinjection, *LssaCA* protein levels in salivary glands decreased dramatically (Fig. 1G [↗](#)). To investigate *LssaCA* secretion, we allowed RSV-free insects to feed on an artificial diet, collecting watery saliva proteins from the diet solution and gel saliva proteins from the feeding parafilm. Western blotting demonstrated that the *LssaCA* protein was secreted as a component of watery saliva (Fig. 1H [↗](#)). Furthermore, by measuring *LssaCA* protein levels in plants after feeding by RSV-free insects, we confirmed that the *LssaCA* proteins can be secreted into plants (Fig. 1I [↗](#)).

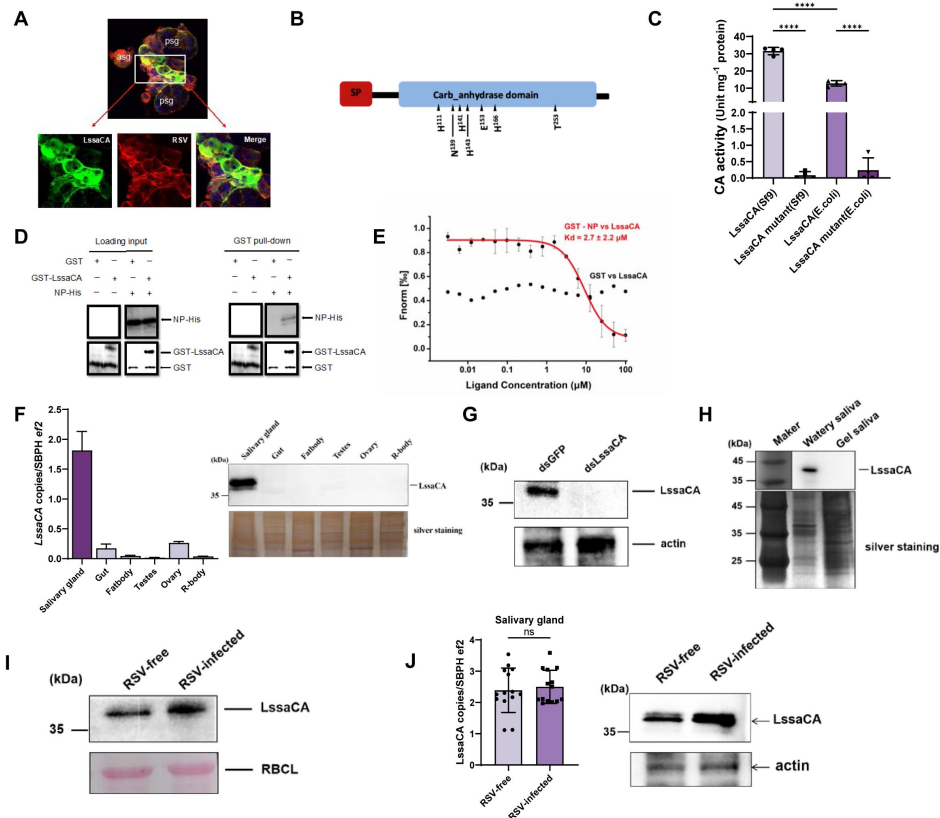


Figure 1.

An *L. striatellus* salivary carbonic anhydrase (LssCA) directly binds to RSV NP in salivary glands

A. Immunofluorescence showing co-localization of RSV (red) and LssCA (green) in *L. striatellus* salivary glands. Images represent three independent experiments with 30 SBPHs analyzed. Scale bar: 20 μ m. psg, principal salivary gland; asg, accessory salivary gland.

B. Schematic of LssCA protein sequence. SP, signal peptide; Carb_anhydrase domain, conserved eukaryotic-type carbonic anhydrase sequence. Triangles indicate seven predicted catalytically active residues.

C. Esterase activity of recombinant LssCA protein expressed in sf9 insect cell and *E. coli* system. In LssCA mutant protein, all seven predicted catalytically active residues were replaced. Mean and SD were calculated from 2 independent enzymatic assays. ****, $P < 0.0001$.

D. GST pull-down assays demonstrating interaction between LssCA (GST-LssCA) and RSV NP (NP-His). GST served as negative control. Anti-GST and anti-His antibodies detected corresponding proteins.

E. MST assay showing binding between LssCA (LssCA-His) and RSV NP (GST-NP). GST served as negative control. Error bars represent SD.

F. Measurement of LssCA tissue-specific expression by RT-qPCR and western blotting. Third-instar RSV-free nymphs were used to measure the gene expression in gut, salivary gland, fatbody, and remaining tissues (R-body) after dissection of the aforementioned tissues. Adult females and adult males were used to measure gene expression in ovaries and testes, respectively. Mean and SD were calculated from three independent experiments with 5 tissue samples each experiment. LssCA protein was detected using anti-LssCA antibodies.

G. Western blot analysis of LssCA protein in RSV-free SBPH salivary glands. dsLssCA treatment significantly reduces target protein levels compared to controls. LssCA protein was detected using anti-LssCA antibodies.

H. Western blot analysis showing LssCA distribution in watery and gel saliva of RSV-free insects. LssCA protein was detected using anti-LssCA antibodies. Total saliva proteins were visualized by silver staining.

I. Western blot detection of LssCA proteins in rice phloem following *L. striatellus* feeding. LssCA protein was detected using anti-LssCA antibodies.

J. Analyses of LssCA levels regulated by RSV infection. Third-instar RSV-free and RSV-infected nymphs were used to measure gene expression (RT-qPCR) and protein accumulation (Western blotting). Mean and SD were calculated from three independent experiments with totally 14 tissue samples each experiment. ns, not significant.

All western blot images are representative of three independent experiments.

We also examined the regulation of *LssaCA* expression by RSV. RSV infection did not significantly influence *LssaCA* mRNA levels in the salivary glands; however, the protein levels were significantly higher in RSV-infected salivary glands (**Fig. 1J** [↗](#)).

In conclusion, these results indicate that when RSV NP is secreted by *L. striatellus*, a salivary-gland-specifically expressed protein, *LssaCA*, can directly bind to RSV and be secreted via saliva.

LssaCA enhances RSV infection in plants

To determine the role of *LssaCA* in RSV infection of rice plants, 36.8 nL ds*LssaCA* at 1 ng/nL was microinjected into the hemocoel of RSV-infected 3rd-instar nymphs to interfere with gene expression. At 2 days post-microinjection (dpi), *LssaCA* mRNA levels were measured by RT-qPCR and compared with the dsGFP control. Results showed that ds*LssaCA* treatment reduced *LssaCA* mRNA levels by 80% (**Fig. 2A** [↗](#)). The two groups of insects were then allowed to feed on healthy rice seedlings (5 insects per rice seedling) for 2 days. RSV NP RNA levels (indicative of RSV titers) in the plants were measured at 14 days post-feeding (dpf), when inoculated viruses had infected the plant host, replicated in plant cells, and viral titers were high enough to be detected by RT-qPCR and western blotting. Compared to rice seedlings fed by dsGFP-treated insects, those fed by ds*LssaCA*-treated insects had significantly reduced RSV titers (**Fig. 2B-C** [↗](#)), indicating that *LssaCA* plays an essential role in mediating RSV infection of rice plants.

To determine at which step *LssaCA* affected RSV infection, we analyzed the influence of *LssaCA*-deficiency on RSV titers during several steps of the RSV transmission and infection process, including virus load in salivary glands and saliva, and initial inoculation levels *in planta*. First, at 2 days after ds*LssaCA* treatment of viruliferous insects, when *LssaCA* was significantly downregulated (**Fig. 2D** [↗](#)), RT-qPCR was used to detect RSV titer in insect salivary glands. Results showed that *LssaCA* deficiency did not reduce RSV titers in this tissue (**Fig. 2E** [↗](#)). Then, *LssaCA*-deficient insects were allowed to feed on an artificial diet, and their saliva was collected for RSV level analysis. Results revealed that *LssaCA*-deficiency did not reduce RSV titer in the secreted saliva (**Fig. 2F** [↗](#)). Finally, a rice plant was fed with 10 viruliferous insects for 24 h, with insects limited to a specific area, and RSV titers were measured immediately after feeding. We found that *LssaCA*-deficiency did not affect RSV initial inoculation levels *in planta* (**Fig. 2G** [↗](#)). We also monitored the influence of *LssaCA* deficiency on the insect feeding behaviors by using Electrical penetration graph (EPG). The EPG waveforms can be classified into three different periods: nonpenetration (NP), penetration (N1, N2, N3) and phloem feeding (N4: N4-a, N4-b) periods. The NP waveform indicates a lack of feeding, the N waveform indicates penetration of mesophyll cells to seek the phloem and the N4 waveform indicates digestion of the sap from the phloem. Results indicated that *LssaCA* deficiency did not affect the insect feeding behaviors (**Fig. S4** [↗](#)). These results collectively suggest that *LssaCA* promotes RSV infection through a mechanism occurring not in insects or during early stages of viral entry in plants, but *in planta* after viral inoculation.

LssaCA protein levels were measured after insect feeding. Plants were fed on by 50 RSV-free insects for 2 days, and then *LssaCA* protein levels were measured at 1 and 3 dpf by western blotting. We found that *LssaCA* was retained for at least 3 days, and protein levels decreased rapidly from 1 dpf to 3 dpf (**Fig. 2H** [↗](#)). We then measured RSV infection levels at 3 dpf to assess the *in-planta* influence after viral inoculation. *LssaCA* expression was knocked down using ds*LssaCA* in RSV-infected 3rd-instar nymphs, and 5 insects were allowed to feed on healthy rice for 2 days. We measured RSV titers in the whole plant shoot at 3 dpf. We found significantly lower RSV titers in the ds*LssaCA*-treatment group at 3 dpf (**Fig. 2I-J** [↗](#)). These results further confirmed the enhancement of RSV's plant infection process by *LssaCA*.

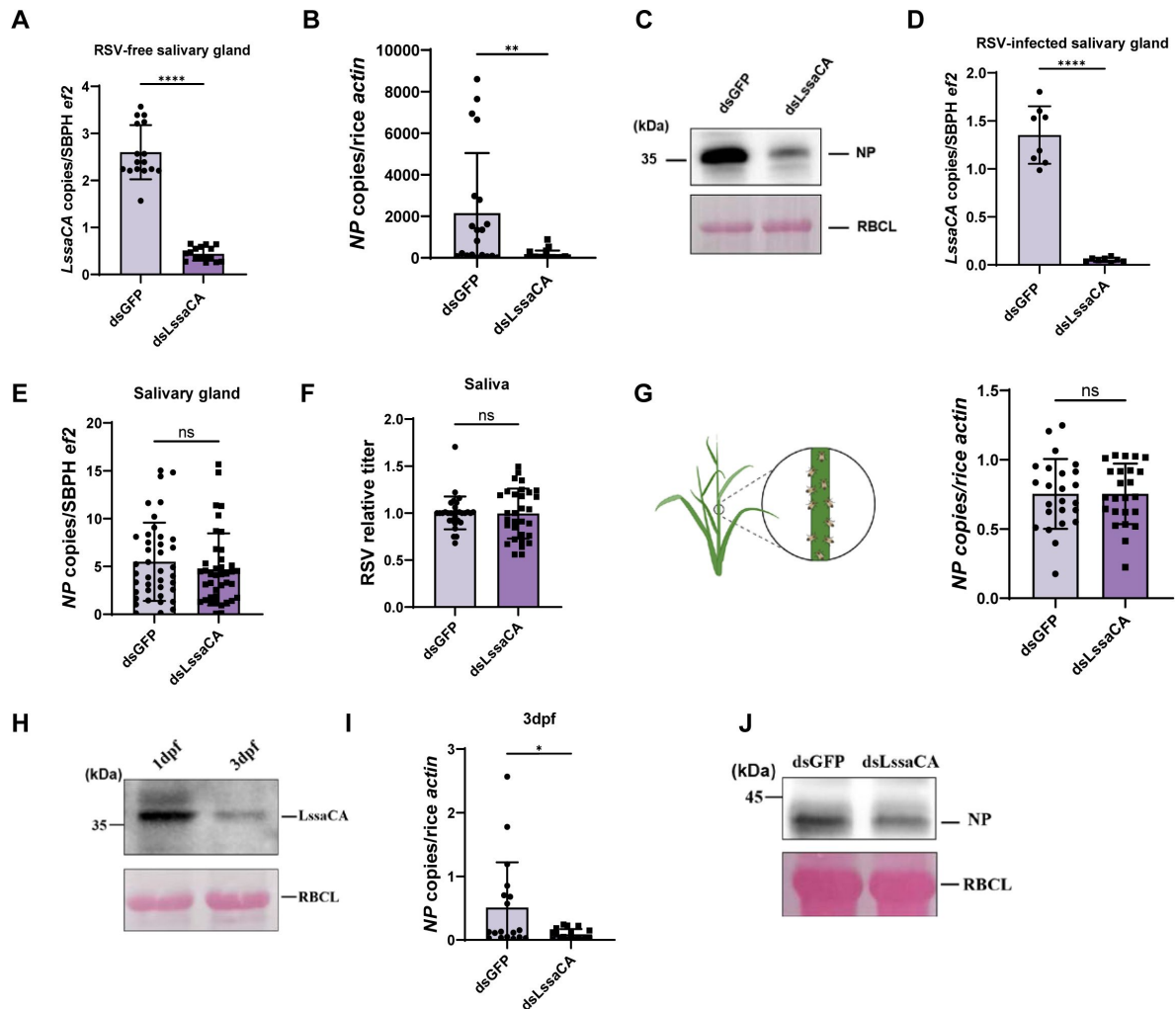


Figure 2.

LssaCA enhances RSV infection in rice plants

A. Gene silencing efficiency determined by RT-qPCR.

B-C. Analysis of RSV infection levels in rice seedlings. Each plant seedling was fed upon by 5 infected insects for 2 days, followed by 14 days of cultivation. RSV levels were determined by RT-qPCR (B) and western blotting (C). RSV NP detected by anti-NP polyclonal antibodies indicated viral titers. Rubisco large subunit (RBCL) served as protein loading control.

D. Gene silencing efficiency determined by RT-qPCR.

E. RT-qPCR analysis of RSV titers in salivary glands. Each dot represents salivary glands from 5 RSV-infected third-instar nymphs.

F. RSV titers in *L. striatellus* saliva determined by RT-qPCR. Each dot represents one saliva sample from 10 insects fed with artificial diet.

G. RT-qPCR analysis of RSV titers in rice. Plants were fed upon by infected insects for 24 h (Left: schematic of *L. striatellus* feeding on specific rice stem sites), with titers assayed immediately post-feeding (Right). Each dot represents an individual rice seedling.

H. Western blot analysis of LssaCA in plants fed upon by 50 RSV-free insects for 2 days and continuously cultured for an additional 1 and 3 days. LssaCA protein was detected using anti-LssaCA antibodies. dpf, days post-feeding.

I-J. RSV infection in plants at 3 dpf. Five RSV-infected third-instar *L. striatellus* nymphs were fed upon healthy rice plants for 2 days, and viral titers in whole plant shoots were measured at 3 dpf by RT-qPCR (I) and western blotting (J). Each dot represents an individual plant. RSV NP detected by anti-NP polyclonal antibodies indicated viral titers. RBCL protein served as a protein loading control.

Mean and SD were calculated from 3 independent microinjection experiments. ns, not significant; *, $P < 0.05$; **, $P < 0.01$; ****, $P < 0.0001$. All western blot images are representative of three independent experiments.

LssaCA interacts with rice thaumatin-like protein to increase its endo- β -1,3-glucanase activity and facilitate RSV infection *in planta*

To investigate the mechanism by which LssaCA facilitates RSV infection in plants, we performed yeast two-hybrid screening to identify rice proteins that interact with LssaCA. A rice thaumatin-like protein (XP_025879846.1), designated as *Oryza sativa* thaumatin-like protein (OsTLP) (**Fig. 3A-B** and **S5, Table S1**), was identified. The open reading frame of OsTLP encodes a polypeptide of 327 amino acids with a thaumatin family (THM) domain from residues 27 to 267. A BLAST algorithm-based analysis revealed sequence identities to TLP 1b. The N-terminal 24 amino acids constitute a secretion signal peptide, suggesting OsTLP is a secreted protein. Pull-down and MST assays were performed to confirm the molecular interaction between OsTLP and LssaCA. Pull-down assays showed that OsTLP and LssaCA have a specific interaction (**Fig. 3C**). The MST assay revealed a K_D of $1.3 \pm 2.2 \mu\text{M}$ for the OsTLP-LssaCA interaction, further confirming the molecular interaction (**Fig. 3D**).

Previous studies have shown that TLP orthologs in various plants, including apples, cherries, tomatoes, and barley, possess endo- β -1,3-glucanase activity that degrades callose (de Jesus-Pires et al., 2020). To verify the endo- β -1,3-glucanase activity of OsTLP, we conducted enzymatic assays using recombinantly expressed OsTLP protein. The purified protein exhibited an average activity of 60 units/mg (**Fig. 3E**). To examine OsTLP activity *in planta*, we constructed transgenic plants overexpressing OsTLP (OsTLP OE) by using the *Japonica* rice cultivar Nipponbare (**Fig. S6A-B**). These transgenic plant showed significantly increased β -1,3-glucanase activity compared to wild-type controls (**Fig. 3F**).

To determine whether the interaction between LssaCA and OsTLP affects enzymatic activity, we pre-incubated OsTLP with either LssaCA or BSA (control) for 2 h before measuring activity. Pre-incubation with LssaCA significantly enhanced OsTLP activity compared to the BSA control (**Fig. 3G**). Since CA family proteins catalyze the conversion of CO_2 and H_2O to HCO_3^- and H^+ , potentially lowering pH, we examined the pH dependence of OsTLP activity. Activity assays conducted across a pH range of 5.0-7.5 revealed that OsTLP activity increased at lower pH values (**Fig. 3H**). However, pH monitoring during OsTLP-LssaCA interaction experiments showed no change from the initial pH of 7.4, indicating that the enhanced enzymatic activity was not due to pH changes but rather to the direct interaction between OsTLP and LssaCA.

Moreover, we measured RSV infection levels in plants with increased OsTLP activity. Viruliferous *L. striatellus* fed on either OsTLP OE plants or wild-type plants for 2 days, and RSV titers were measured at 14 dpf. RT-qPCR analyses indicated that compared with wild-type plants, plants overexpressing OsTLP had significantly higher RSV titers (**Fig. 3I**), indicating that OsTLP facilitates RSV infection *in planta*.

Collectively, these results suggest that LssaCA may promote RSV infection by enhancing OsTLP's callose-degrading activity.

RSV NP-LssaCA-OsTLP tripartite interaction further enhances OsTLP enzymatic activity

Given that LssaCA interacts with both RSV NP and OsTLP, we investigated the possibility of a tripartite interaction. Through pull-down assays, we discovered that when LssaCA was pre-incubated with NP to form an NP-LssaCA complex, this complex retained the ability to bind to OsTLP (**Fig. 4A**). This result demonstrates a tripartite interaction among these three proteins.

We then explored how the tripartite interaction affects OsTLP enzymatic activity. OsTLP was pre-incubated with either the LssaCA-NP complex or LssaCA alone as a control. After 2 h of incubation, enzyme activity was measured. The results showed that pre-incubation with the LssaCA-NP

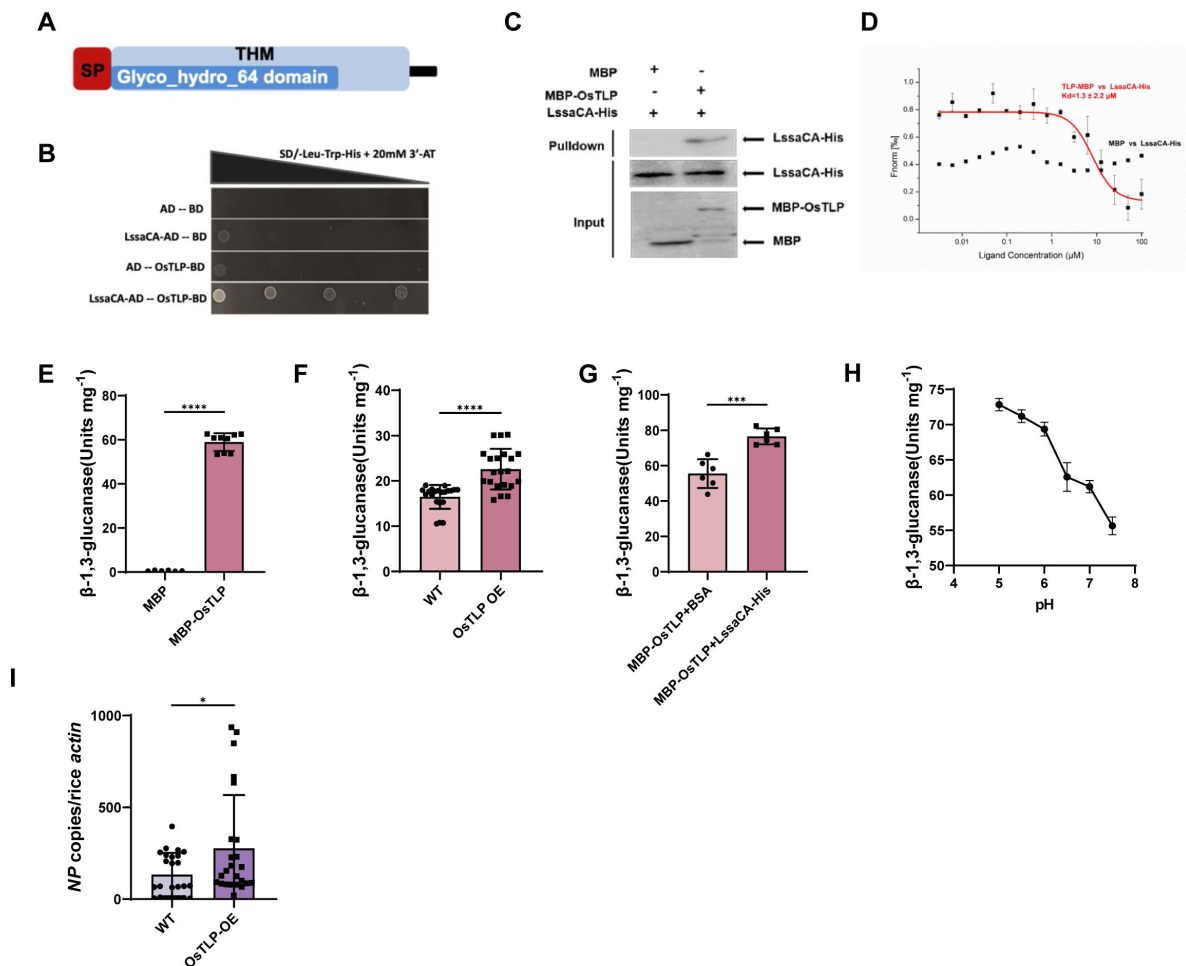


Figure 3.

LssaCA interacts with rice thaumatin-like protein (OsTLP) to enhance its endo-β-1,3-glucanase activity

- A. Schematic diagram of OsTLP protein sequence. SP, signal peptide; THM, conserved thaumatin family protein sequence. Glyco_hydro_64 domain, glycoside hydrolases of family 64.
- B. Yeast two-hybrid assay showing interaction between OsTLP and LssaCA. SD, synthetically defined medium; Leu, Leucine; Trp, Tryptophan; His, Histidine; 3'AT, 3-amino-1,2,4-triazole; AD, transcription activation domain; BD, DNA-binding domain.
- C. Pull-down assays showing interactions between LssaCA (LssaCA-His) and OsTLP (MBP-OsTLP). MBP served as a negative control. Anti-His and anti-MBP polyclonal antibodies were used for protein detection. Western blot images are representative of three independent experiments.
- D. MST assay showing molecular interactions between LssaCA (LssaCA-His) and OsTLP (MBP-OsTLP). MBP served as a negative control. Error bars represent SD.
- E. Endo-β-1,3-glucanase activity of purified recombinant OsTLP protein expressed as MBP-fusion protein. MBP was used for normalization.
- F. Endo-β-1,3-glucanase activity in OsTLP-overexpressing transgenic plants (OsTLP OE) or wild-type plants (WT). Plant total soluble proteins were used for enzymatic activity assays. Units mg^{-1} means units per mg of plant total soluble proteins.
- G. Regulation of OsTLP enzymatic activity by LssaCA binding.
- H. OsTLP enzymatic activity measured at pH 5.0, 5.5, 6.0, 6.5, 7.0 and 7.5 to assess pH dependence.
- I. RT-qPCR analysis of RSV infection in OsTLP OE and WT plants. RSV titers (NP copy number) were measured at 14 dpf. ns, not significant; *, $P < 0.05$; ***, $P < 0.001$; ****, $P < 0.0001$.

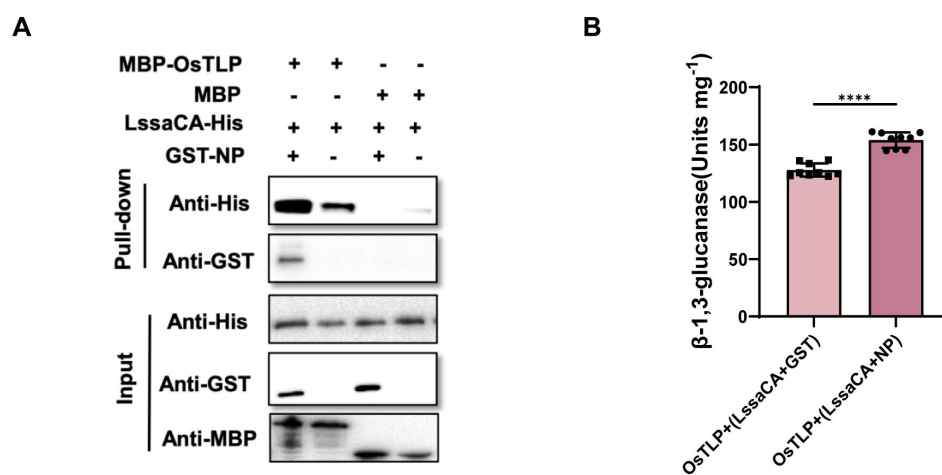


Figure 4.

RSV NP-LssaCA-OsTLP tripartite interaction further enhances OsTLP enzymatic activity

A. Pull-down assays showing interactions between OsTLP (MBP-OsTLP) and LssaCA-NP complex. LssaCA was expressed with a His tag, and RSV NP was expressed as a GST-fusion protein. LssaCA and NP were pre-incubated before co-incubation with OsTLP. Anti-His, anti-GST, and anti-MBP antibodies were used to detect corresponding proteins.

B. Endo-β-1,3-glucanase activity of OsTLP after co-incubation with LssaCA (LssaCA + GST) and LssaCA-NP complex (LssaCA + NP). LssaCA was pre-incubated with either GST or NP before addition to OsTLP. ****, $P < 0.0001$.

complex significantly increased OsTLP activity compared to the LssaCA control (Fig. 4B). This indicates that stronger activation of β -1,3-glucanase activity is achieved through the NP-LssaCA-OsTLP tripartite interaction.

LssaCA inhibits insect-feeding and RSV-infection induced callose deposition in rice plants

Callose deposition on sieve plates and plasmodesmata serves as an critical defense mechanism in plants, reducing insect feeding, viral spread, and infection severity (Li et al., 2012; Zavaliev et al., 2011). Given that the LssaCA or LssaCA-NP complex enhances OsTLP's β -1,3-glucanase activity and potentially promotes callose degradation, we examined callose regulation during feeding by both RSV-free and RSV-infected *L. striatellus*. To assess feeding-induced callose deposition, we exposed rice seedlings to either RSV-free or RSV-infected *L. striatellus* for 24 h and analyzed callose levels around the feeding sites. ELISA assays showed that while feeding by RSV-free insects increased callose concentration, feeding by RSV-infected insects triggered an even greater increase (Fig. 5A). Consistent with these findings, microscopic analysis revealed enhanced callose deposition in phloem cells of plants fed upon by RSV-infected *L. striatellus* compared to those fed by RSV-free insects (Fig. 5B-C and S7). To determine whether this response was localized, we measured callose levels in tissues distant from the feeding sites. No significant differences in callose deposition were observed between control plants and the non-feeding areas (Fig. S8A), indicating that callose accumulation was confined to feeding sites and their immediate vicinity.

To evaluate the molecular mechanisms underlying the induced callose deposition, we analyzed transcript levels of callose synthase genes using RT-qPCR. In agreement with the callose deposition patterns, feeding by RSV-free insects evaluated the expression of callose synthase genes, including *gsl3*, *gsl4*, *gsl5*, and *gsl10*, while RSV-infection further enhanced their upregulation (Fig. 5D-G and S8B-E). These results further support that both insect feeding and RSV infection promote callose deposition. Notably, the transcriptional levels of these genes remained unchanged in non-feeding areas (Fig. S8B-E). We also measured the *OsTLP* gene expression and found that neither SBPH feeding nor RSV infection significantly induced *OsTLP* expression (Fig. 5H).

To investigate whether LssaCA influences callose deposition induced by insect-transmitted RSV, we downregulated *LssaCA* in RSV-infected *L. striatellus* using dsRNA microinjection. At 2 dpi, when *LssaCA* expression was significantly reduced, we allowed the insects to feed on healthy plants for 24 h before measuring callose deposition. Our results showed that plants fed upon by *LssaCA*-deficient insects accumulated more callose compared to those fed upon by control insects (dsGFP treatment) (Fig. 5I-K). These results indicate that LssaCA inhibits callose deposition.

In conclusion, these results demonstrate that both *L. striatellus* feeding and RSV infection induce callose deposition, which can be suppressed by LssaCA.

LssaCA mediates *in planta* callose degradation by enhancing OsTLP enzymatic activity

To confirm whether OsTLP degrades the deposited callose induced by insect-transmitted RSV *in planta*, we constructed *ostlp* mutant plants (Fig. S6C). Both ELISA assays and microscopic analysis showed that compared to wild-type plants, *ostlp* plants exhibited significantly higher callose levels (Fig. 6A-B).

Given that LssaCA mediates callose degradation by enhancing OsTLP enzymatic activity, we hypothesized that when OsTLP is knocked out in plant, LssaCA would not affect callose levels induced by RSV infection. To test this, we allowed dsLssaCA-treated viruliferous *L. striatellus* or dsGFP-treated control insects to feed on *ostlp* seedlings. ELISA assays indicated that LssaCA deficiency did not significantly change callose accumulation (Fig. 6C). Microscopic analysis

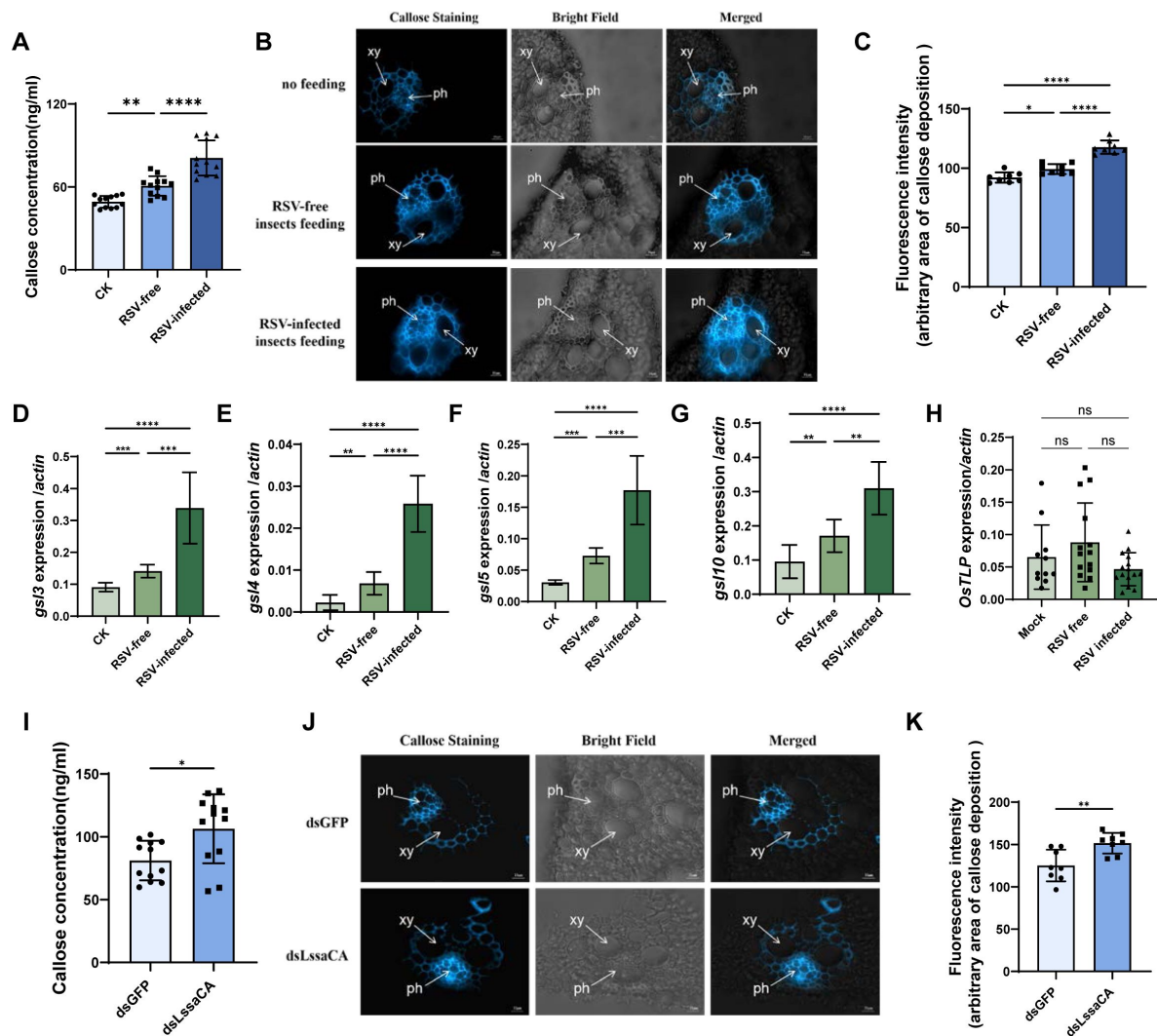


Figure 5.

LssaCA inhibits *L. striatellus*-transmitted RSV-induced callose deposition in rice plants

A. Callose concentration quantified by ELISA in leaves of rice plants fed on by RSV-free or RSV-infected *L. striatellus*. Non-fed plants served as the control (CK).

B. Bright blue fluorescence of cross-sections showing callose deposition at feeding sites. Samples were prepared from the leaf phloem of plants that were not fed or were fed on by RSV-free or RSV-infected *L. striatellus*. Thin sections were stained with 0.1% aniline blue at 24 h post-feeding and examined under a fluorescence microscope. xy, xylem; ph, phloem. Scale bars: 10 μ m.

C. Average fluorescence intensity of callose deposition measured using ImageJ. Eight to ten random sites per sample were selected for intensity quantification.

D–H. Transcript levels of callose synthase genes (*gls3*, *gls4*, *gls5*, *gls10*) and *OsTLP* determined by RT-qPCR. Insects were allowed to feed on rice plants for 24 h prior to RNA extraction.

I. Callose concentration quantified by ELISA in leaves of rice plants fed on by dsGFP- or dsLssaCA-treated RSV-infected *L. striatellus*.

J. Bright blue fluorescence of cross-sections showing callose deposition at feeding sites. Samples prepared from the leaf phloem of plants fed on by dsGFP- or dsLssaCA-treated RSV-infected *L. striatellus*. Scale bars: 20 μ m.

K. Average fluorescence intensity of callose deposition measured using ImageJ. Eight to ten random sites per sample were selected for intensity quantification.

Mean and SD were calculated from three independent experiments. ns, not significant; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.

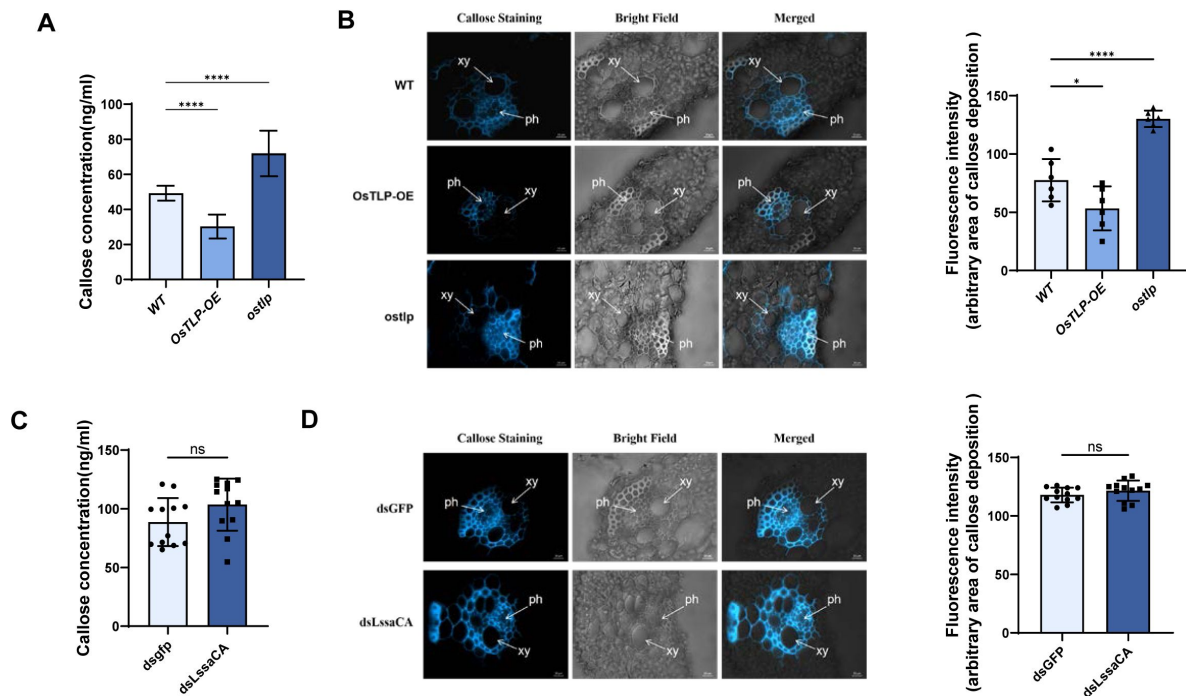


Figure 6.

LssaCA mediates *in planta* callose degradation by enhancing OsTLP enzymatic activity

A. Callose concentration quantified by ELISA in leaves of wild-type (WT), *OsTLP* overexpression (*OsTLP* OE), and *OsTLP* mutant (*ostlp*) plants.

B. Bright blue fluorescence of cross-sections showing callose deposition in leaf phloem. Plants were not fed on by insects. Scale bars: 10 μ m. Average fluorescence intensity of callose deposition was measured using ImageJ. Eight to ten random sites per sample were selected for intensity quantification. xy, xylem; ph, phloem.

C. Callose concentration quantified by ELISA in leaves of *ostlp* plants fed on by dsGFP- or dsLssaCA-treated RSV-infected *L. striatellus*.

D. Bright blue fluorescence of cross-sections showing callose deposition at feeding sites. Samples were prepared from the leaf phloem of *ostlp* plants fed on by dsGFP- or dsLssaCA-treated RSV-infected *L. striatellus*. Scale bars: 10 μ m. Average fluorescence intensity was quantified using ImageJ. Eight to ten random sites per sample were selected for intensity quantification.

Mean and SD were calculated from three independent experiments. ns, not significant; *, $p < 0.05$. ****, $p < 0.0001$.

confirmed higher callose deposition in phloem cells of *ostlp* seedlings, which remained unaffected by LssaCA deficiency (**Fig. 6D**). These results indicate that OsTLP is required *in planta* for LssaCA-mediated effects on callose deposition.

Discussion

Insects are known to be vectors of many plant viruses, and some of these plant viruses rely entirely on insect vectors for their plant-to-plant transmission. Insect salivary proteins have been reported to facilitate viral infection of plants through various mechanisms, such as breaking down plant cell walls, inhibiting plant immune defenses and callose deposition, directly binding to viruses to prevent their detection by plant immune systems, and more. In this study, we found that RSV infection of rice plants triggers the deposition of callose. However, this process is reversed by a tripartite interaction involving the central component of an insect salivary protein LssaCA, which subsequently binds to RSV in insect salivary glands and then OsTLP in the plants. The activity of OsTLP's β -1,3-glucanase is significantly increased when this tripartite interaction occurs; leading to the degradation of callose and thus allowing for efficient RSV infection (**Fig. 7**).

Callose is a complex polysaccharide, and its deposition in plant phloem plays an important role in defending against insect attacks and viral spread (Zavaliev et al., 2011). When attacked by piercing insects or viruses, plants activate callose deposition on the sieve plates to occlude the flow of phloem sap, thus discouraging feeding and reducing viral spread. This accumulation of callose in plant phloem can be switched on or off through the action of callose synthases and hydrolases (Li et al., 2012; Shangguan et al., 2018). Studies have shown that expression levels of these glucanase enzymes differ between resistant and susceptible plant species. This suggests that some plants may be more successful at resisting infection due to high levels of glucanase enzymes in their tissues, which allow them to more quickly and effectively form callose barriers against insects and viruses (Hao et al., 2008). Some insects have evolved β -1,3-glucanases enzymes that can degrade the deposited callose, thus enabling them to bypass the plant's defense mechanisms and continue feeding (Bucher et al., 2001; Hao et al., 2008). In this study, we demonstrate a novel mechanism by which an insect saliva protein binds to a plant β -1,3-glucanases protein, activating the enzymatic activity and promoting callose degradation.

In this study, we found that plants fed on by viruliferous insects exhibited higher callose deposition than those fed on by uninfected insects (**Fig. 5A–C**), suggesting that viral infection *via* insect saliva may constitute a dual stress to the plant, inducing enhanced callose deposition that must be overcome for successful insect feeding and viral spread. To investigate the underlying mechanism, we examined the tripartite interaction between LssaCA-NP-OsTLP. We found that this interaction induced significantly higher β -1,3-glucanase activity than the LssaCA-OsTLP interaction alone, suggesting that SBPH and RSV have evolved mechanisms to enhance β -1,3-glucanase activity in order to bypass the callose-mediated defense deposition and facilitate successful viral spread.

Carbonic anhydrase catalyzes the reversible hydration of carbon dioxide according to the reaction $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}^+$. This enzymatic activity is essential for maintaining pH homeostasis in biological systems by regulating the balance between CO_2 , bicarbonate, and protons. The importance of carbonic anhydrases in pH regulation is exemplified in mosquito larvae, which utilize an extremely alkaline digestive strategy (pH 10.5–11) that is relatively rare in nature, with carbonic anhydrases serving as the central enzymatic system for generating and maintaining bicarbonate/carbonate buffering in the midgut lumen (Linser et al., 2009). Similarly, carbonic anhydrases in arthropod saliva can significantly impact host-pathogen interactions through pH modulation. A recent study elucidated how aphid salivary carbonic anhydrase II (CA-II) facilitates virus transmission, with the enzyme showing significantly higher expression levels in winged

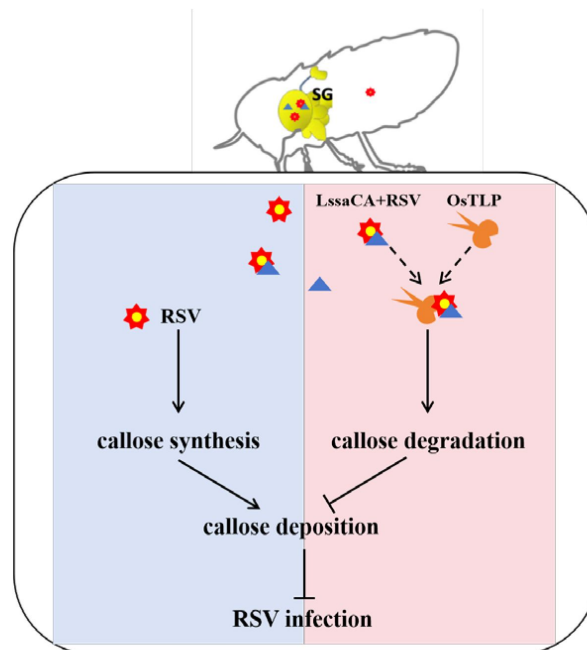


Figure 7.

Proposed model of RSV NP-LssaCA-OsTLP tripartite interaction facilitating callose degradation and promoting RSV infection.

In *L. striatellus* salivary glands (SG), RSV (red star) binds to LssaCA proteins (blue triangle) via the nucleocapsid protein (NP). Upon delivery into the rice phloem, RSV infection triggers callose synthesis, leading to callose deposition that restricts viral spread. However, LssaCA binds to OsTLP (yellow scissors), enhancing its β -1,3-glucanase activity and promoting callose degradation. The RSV NP-LssaCA complex formed in the insect's SG can still interact with OsTLP *in planta*, resulting in significantly higher OsTLP β -1,3-glucanase activity. This enhanced enzymatic activity efficiently degrades callose, thereby facilitating RSV infection.

morphs compared to wingless counterparts. When secreted into the plant apoplastic space, CA-II catalyzes H^+ accumulation, resulting in apoplastic acidification from 5.5 to 5.0. This pH reduction enhances polygalacturonase activity, thereby promoting the degradation of demethylesterified homogalacturonan components. Consequently, plants respond by accelerating vesicle trafficking to transport pectin for cell wall reinforcement, a process that simultaneously facilitates viral translocation from the endomembrane system to the apoplastic compartment (Guo et al., 2023). In contrast to these pH-dependent mechanisms, our study demonstrated that LssaCA's biological function in mediating RSV infection is, if not completely, at least partially independent of its enzymatic activity. We found that although OsTLP enzymatic activity is highly pH-dependent and exhibits increased enzymatic activity as pH decreases from 7.5 to 5.0 (Fig. 3H), the LssaCA-OsTLP interaction occurring at pH 7.4 significantly enhanced OsTLP enzymatic activity without any change in buffer pH (Fig. 3G). These results demonstrate the crucial importance of LssaCA-OsTLP protein interactions, rather than enzymatic activity alone, in mediating RSV infection. While the exact role of LssaCA enzymatic activity in aiding RSV infection remains unclear, the local low-pH plant environment would enhance OsTLP enzymatic activity, suggesting potential synergistic effects between enzymatic and non-enzymatic functions.

In conclusion, this study elucidated a tripartite interaction among a plant virus, an insect saliva protein, and a host phloem β -1,3-glucanase. This interaction inhibited the plant defense of callose deposition and enabled viral phloem transmission. This study provides new insights into the tripartite virus-insect vector-plant interaction, which is relevant to many agriculturally important plant arboviruses whose transmission is facilitated via insect saliva proteins.

Materials and methods

Viruses, insects, host plants, and antibodies

The RSV-free and RSV-infected *L. striatellus* individuals used in this study were originally captured in Jiangsu Province, China, and were maintained in our laboratory. All plants used to rear *L. striatellus* were grown in a growth incubator (14.5 cm height, 3.2 cm radius) at 25°C under a 16-h light/8-h dark photoperiod. Insects were transferred to fresh seedlings every 15 days to ensure adequate nutrition and insect activity.

RSV ribonucleoproteins (RNPs) were purified from RSV-infected rice plants. These purified RNPs were subsequently used as the antigen to immunize rabbits for the production of an RSV-specific antibody. The LssaCA polyclonal antibody was produced using recombinantly expressed LssaCA protein (with a His-tag at the C-terminus) as the antigen. Anti-His, anti-GST, and anti-MBP antibodies were purchased from ABclonal.

Tissue collection

Insect dissection and tissue collection were performed using previously described procedures (Huo et al., 2018). In brief, insects were anesthetized at 4°C for 10 min, and the forelegs were severed at the coxa-trochanter joint using forceps. The hemolymph was expelled and drawn to the tip of clean forceps. The insects were then dissected from the abdomen in pre-chilled PBS buffer. The fat body was collected with forceps, aided by the tension of the liquid. Tissues, including the guts, salivary glands, and ovaries, were washed three times in PBS to remove contaminants from the hemolymph.

Bioinformatic analysis

The signal peptide and transmembrane helix were predicted using tools at the SignalP 5.0 server (<https://services.healthtech.dtu.dk/service.php?SignalP-5.0>) and the TMHMM-2.0 server (<https://services.healthtech.dtu.dk/service.php?TMHMM-2.0>). Homology searches of the protein sequence of

LssaCA and OsTLP were conducted using the BLAST algorithm at the National Center for Biotechnology Information (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). Conserved domains were predicted with SMART (<http://smart.embl-heidelberg.de/>).

Collection of watery and gel saliva

Saliva was collected as described previously (Huo et al., 2022). Watery saliva was obtained by centrifuging the collected artificial diet at 10,000 ×g for 10 min to remove the solid gel. To collect gel saliva proteins, the membrane was washed twice with PBS. Sheathes adhering to the parafilm were denatured and solubilized in a solution consisting of 8 M urea, 4% v/v CHAPS, 0.1% w/v sodium dodecyl sulfate (SDS), and 2% v/v DTT. The parafilm was incubated on an orbital shaker at room temperature for 45 min. Then, the denatured gel saliva was concentrated to a final volume of 100 µL. Saliva samples were collected from 2,000 insects for silver staining and western blotting analyses.

Measurement of carbonic anhydrase activity

DNA fragments encoding full-length LssaCA and a LssaCA mutant (H111D, N139H, H141D, H143D, E153H, H166D, and T253E) were amplified by PCR. The resulting products were purified and cloned into the vector pET-30a(+) and pFastBac1 vectors to generate His-tagged recombinant proteins. The pET-30a(+)-LssaCA and pET-30a(+)-LssaCA(mutant) plasmids were transformed into *E. coli* for protein expression. In parallel, the pFastBac1-LssaCA and the pFastBac1-LssaCA(mutant) plasmids were transformed into *E. coli* DH10Bac for transposition into the bacmid. Recombinant bacmids were then transfected into *Spodoptera frugiperda* (Sf9) cells using Cellfectin II Reagent (Invitrogen, USA). Recombinant proteins expressed in both systems were purified using Ni Sepharose resin (GE Healthcare, Uppsala, Sweden).

Carbonic anhydrase activity was assessed by measuring esterase activity using *p*-nitroethyl acetate (*p*-NPA) as the substrate. LssaCA and its mutant were expressed in *E. coli* and Sf9 cells with C-terminal His-tag, and the purified proteins were diluted in PBS buffer (pH 7.0) to a final concentration of 1 mg/mL. Reaction conditions followed the method described by Verpoorte et al. (1967), with minor modifications. Briefly, 1 mL of freshly prepared 3 mM *p*-NPA (dissolved in acetone) was added to 1.9 mL of 100 mM PBS (pH 7.0), followed by 0.1 mL of LssaCA enzyme solution (0.5 mg/mL in 100 mM PBS). Absorbance at 384 nm was measured at the start of the reaction and after 3 minutes. One unit (U) of enzyme activity was defined as the amount of enzyme required to hydrolyze 1 µmol of *p*-NPA per minute at pH 7.0 and 25 °C. A blank control consisted of 2 mL of 100 mM PBS (pH 7.0) mixed with 1 mL of 3 mM *p*-NPA.

Immunofluorescence assays

Immunofluorescence assays (IFA) were performed to determine the localization of RSV and LssaCA in *L. striatellus* salivary glands. The dissected salivary gland samples were fixed in 4% v/v paraformaldehyde at 4°C overnight and then permeabilized in 2% v/v Triton X-100 for 2 h. The samples were incubated with rabbit anti-RSV serum and mouse anti-LssaCA serum (1:250 dilution in PBST/FBS: PBS containing 2% v/v Triton X-100 and 2% fetal bovine serum) for 1 h at room temperature. After washing twice with PBST, the samples were incubated with 1:200 diluted Alexa Fluor 488-conjugated goat anti-mouse antibody and Alexa Fluor 594-conjugated goat anti-rabbit antibody (Invitrogen, USA) for 1 h. After washing three times with PBST, the samples were examined under a Leica TCS SP8 microscope (Leica, Wetzlar, Germany) with the following settings: 1) Alexa Fluor 488 channel: excitation at 488 nm using an argon laser; emission detected with a 493-553 nm bandpass filter. Alexa Fluor 594 channel: excitation at 594 nm using a HeNe laser; excitation detected with a 600-650 nm bandpass filter.

Evaluation of callose deposition in rice plants

Five insects were allowed to feed for 24 h on a 2 × 2 cm leaf area of 10-d-old rice plants. After removal of the insects, leaf tissues were collected from both the feeding site and an adjacent undamaged area located 1 cm away (non-fed site). Samples were immersed in 5 mL alcoholic lactophenol solution (1 volume of phenol:glycerol:lactic acid:water in a 1:1:1:1 ratio, mixed with 2 volumes of ethanol), evacuated with a syringe for 15 min, and then incubated at 65°C for 30 min until the chlorophyll was completely cleared. Cleared leaf samples were rinsed with 50% v/v ethanol, then with water, and then stained in 150 mM K₂HPO₄ (pH 9.5) containing 0.01% w/v aniline blue for 30 min in the dark. Stained leaf samples were sectioned and mounted in 50% v/v glycerol on slides. The slides were examined under a Zeiss observer Z1 immunofluorescence microscope (Carl Zeiss MicroImaging GmbH, Gottingen, Germany) with excitation at 405 nm. Fluorescence intensity was quantified using ImageJ software. For ImageJ analysis, images were converted to 8-bit grayscale and adjusted for brightness. Phloem-specific regions of interest (ROIs) were outlined using the freehand selection tools. A standardized fluorescence threshold (set at >2× the mean background intensity) was applied to isolate callose signals. Integrated density and area values were used to calculate normalized fluorescence intensity. Identical exposure times were maintained across all independent experiments.

Callose concentrations were quantified using a plant callose ELISA kit (Catalog No. JL46407; Jonln, Shanghai, China), following the manufacturer's instruction. Leaf tissues from feeding and non-fed sites were homogenized in PBS buffer (pH 7.4), and the homogenate was centrifuged at 500–1000 × g for 20 min. The supernatant was collected, and absorbance at 450 nm was measured. Callose concentrations were calculated using a standard curve.

RNA interference

We conducted RNAi analyses to determine the function of *LssaCA* in mediating RSV infection and regulating callose deposition. The *LssaCA*-specific gene fragment was PCR-amplified with the primer pair *LssaCA*-T7F/*LssaCA*-T7R (Table S1). Then, dsRNA was synthesized using a commercial kit (T7 RiboMAX Express RNAi System, Promega, Madison, WI, USA) and purified by phenol:chloroform extraction and isopropanol precipitation. Finally, 36.8 nL dsRNA at 1 ng/nL was delivered into the insect hemocoel for gene silencing. Microinjection was performed using a glass needle through a Nanoliter 2000 (World Precision Instruments, Sarasota, FL, USA). The negative control, GFP dsRNA, was synthesized and microinjected following the same protocol.

RT-qPCR

Gene transcript levels were determined by RT-qPCR. The RNA was extracted from *L. striatellus* tissues using a RNeasy Mini Kit (Qiagen, Hilden, Germany), and from plant leaves using TRIzol. Reverse transcriptional PCR (iScript cDNA Synthesis Kit, Bio-Rad, Hercules, CA, USA) and SYBR-Green-based qPCR (SYBR Green Supermix, Bio-Rad) was performed according to the manufacturer's protocols. The primer pair used to amplify *LssaCA* was *LssaCA*-q-F/*LssaCA*-q-R (Table S1). Viral RNA copies were measured using the primers NP-q-F/NP-q-R (Table S1). *L. striatellus* translation elongation factor 2 (*EF2*) and rice *actin* were amplified as internal controls to ensure equal loading of cDNA isolated from different samples. The primer pairs *EF2*-q-F/*EF2*-q-R and *actin*-q-F/*actin*-q-R (Table S1) were used to amplify *EF2* and *actin*, respectively. Water was used as a negative control.

Yeast two-hybrid assays

Yeast two-hybrid screening was performed to identify rice proteins interacting with *LssaCA*. A high-complexity rice cDNA library was fused to *Gal4 AD* and transformed into the yeast strain Y187. The *LssaCA* gene lacking the signal-peptide sequence (*LssaCA*^{SP}) was fused to *Gal4 DNA-BD* and transformed into the yeast strain Y2HGold. The two yeast strains were co-cultured overnight

to allow mating to create diploids. After library screening, positive clones were selected on quadruple dropout medium: SD/-Ade/-His/-Leu/-Trp supplemented with X-a-Gal and aureobasidin A (QDO/X/A), and prey plasmids were isolated for sequencing. To further confirm the interaction between LssaCA and OsTLP, *LssaCA*^{SP} was cloned into the prey vector pDB-Leu, and the *OsTLP* gene lacking the signal-peptide sequence (*OsTLP*^{SP}) was cloned into the bait vector pPC86. The two plasmids were co-transformed into the yeast strain Mav203 and the transformants were selected on SD/-His/-Leu/-Trp + 20 mM 3-AT medium.

Pull-down assays

Pull-down assays were performed to determine the molecular interactions between LssaCA and RSV NP, LssaCA and OsTLP, and OsTLP and the LssaCA-NP complex. The interaction between LssaCA and RSV NP was determined using GST pull-down analyses as described previously (Huo et al., 2014). Briefly, *LssaCA*^{SP} was cloned into the pGEX-3x vector and LssaCA was expressed as a GST-fusion protein (GST-LssaCA). The NP gene was cloned into the pET-30a (+) expression vector and the protein was expressed with an N-terminal His-tag (NP-His). The GST-LssaCA construct was expressed in, and purified from, *E. coli* strain BL21. The GST-LssaCA protein or the GST control protein was bound to Glutathione Sepharose beads (Cytiva, Marlborough, MA, USA), then NP-His protein was added to the beads. The bead-bound proteins were separated by SDS-PAGE and detected by western blotting with antibodies against GST or His-tag.

Next, MBP pull-down analyses were performed to determine the interaction between OsTLP and LssaCA. The *OsTLP*^{SP} sequence was cloned into pMAL-p2X to produce the MBP-fusion protein (MBP-OsTLP). The MBP-OsTLP protein or the MBP control protein was bound to amylose resin (New England Biolabs, Beverly, MA, USA), and then the LssaCA-His protein was added to the beads. To determine the interaction between OsTLP and the LssaCA-NP complex, the LssaCA-His and GST-NP proteins were preincubated before being added to the beads. The bead-bound proteins were separated by SDS-PAGE and detected by western blotting with the MBP antibody (Cell Signaling Technology, Danvers, MA, USA), His-tag antibody (Tiangen, Beijing, China) and GST antibody (Tiangen).

Microscale thermophoresis (MST) assays

We conducted MST assays to detect the interactions between LssaCA and NP, and between LssaCA and OsTLP. First, 10 μ M purified LssaCA^{SP} protein (His-fusion protein) was labelled with a Monolith NT Protein Labeling Kit RED-NHS (Nano Temper Technologies GmbH, München, Germany) using red fluorescent dye NT-647 N-hydroxysuccinimide (amine-reactive), according to the manufacturer's instructions. The binding assays were performed on a Monolith NT.115 Microscale Thermophoresis instrument (Nano Temper Technologies GmbH) using capillaries treated using the standard method. The labeled protein LssaCA^{SP} was added to serially diluted NP (GST-NP, GST was used as a negative control) or *OsTLP*^{SP} (MBP-fusion protein, MBP was used as a negative control). The initial concentration of each protein was 40 μ M with 0.1% (v/v) Tween 20. The KD Fit function of NanoTemper Analysis software version 1.2.214.1 was used for curve fitting and to calculate the dissociation constant (Kd).

β -1,3-glucanase activity assays

β -1,3-glucanase hydrolyzes laminarin by cleaving β -1,3-glucoside bond, thereby generating sugar termini. Enzyme activity was quantified by measuring the rate of reducing sugars production. One unit (U) of β -1,3-glucanase activity was defined as the amount of enzyme that hydrolyses laminarin to generate 1 μ g reducing sugars per minute.

The β -1,3-glucanase activity of OsTLP was measured using a beta-1,3-glucanase microplate assay kit (Cat. ASK1028, Bioworld, Bloomington, MN, USA) according to the manufacturer's instructions. To determine the enzymatic activity of recombinant OsTLP proteins, MBP-fused *OsTLP*^{SP} protein

and MBP control protein were adjusted to 0.5 mg/mL. The MBP-OsTLP^{SP} fusion protein was equilibrated in buffers of varying pH for the assay. To determine the enzymatic activity of OsTLP expressed *in planta*, 0.1 g of rice leaf tissue from either OsTLP over-expressing or wild-type plants was homogenized with 1 mL of assay buffer on ice. The homogenate was centrifuged at 12,000g at 4°C for 10 min, and the resulting supernatant was transferred into a new tube and kept on ice until analysis.

For the assay, 50 µL of the protein solution was mixed with laminarin substrate and incubated at 37°C for 30 min. The reaction was terminated by incubating the reaction tube in a boiling water bath for 10 min. The reaction supernatant was transferred into a microplate and incubated at 90°C for 10 min before measuring the absorbance at 540 nm.

Generation of transgenic plants

The full-length *OsTLP* gene was PCR-amplified with the primer pair TLP-F/TLP-R (Table S2) and cloned into the plant expression vector pCambia1300 to produce the plasmid pCambia1300-OsTLP. This construct was transformed into seedlings of the *Japonica* rice cultivar Nipponbare through *Agrobacterium*-mediated transformation, following the protocol described by Hiei et al. (1994) [\[18\]](#). The *ostlp* knockout line was generated using CRISPR-Cas9 technology according to established protocols. Thirty independent T₀ transformants were obtained and screened by germinating seeds from T₂ lines on medium containing 50 µg/ml hygromycin. Hygromycin-resistant homozygous lines were selected for further experiments. Knockout events were confirmed by sequencing with specific primers (Fig S6C). Seeds from homozygous lines transformed with the vector pCambia1300-eGFP were used as negative control.

Statistical analyses

Graphs were generated, and statistical analyses were conducted using Prism 9.0 software (GraphPad Software, San Diego, CA, USA). Data are presented as mean ± standard deviation (SD) from two or three independent experiments. Statistical significance between two groups was assessed using an unpaired Student's t-test, while comparisons among three or more groups were performed using one-way analysis of variance (ANOVA).

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Additional information

Author contributions

Conceptualization, Z.L.L., H.Y., Z.J.; Methodology, H.Y., Z.J. and Y.J.; Investigation, Z.J., Y.J., M.X.Y., H.Y.; Writing-Original Draft, H.Y. and Z.J.; Writing-Review & Editing, Z.L.L. and H.Y.; Funding Acquisition, Z.L.L., H.Y. and F.R.X.; Supervision, Z.L.L., and F.R.X.

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Additional files

Supplemental figures and tables [🔗](#)

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Reviewer #1 (Public review):

In this study, the authors identified an insect salivary protein LssaCA participating viral initial infection in plant host. LssaCA directly bond to RSV nucleocapsid protein and then interacted with a rice OsTLP that possessed endo- β -1,3-glucanase activity to enhance OsTLP enzymatic activity and degrade callose caused by insects feeding. The manuscript suffers from fundamental logical issues, making its central narrative highly unconvincing.

(1) These results suggested that LssaCA promoted RSV infection through a mechanism occurring not in insects or during early stages of viral entry in plants, but in planta after viral inoculation. As we all know that callose deposition affects the feeding of piercing-sucking insects and viral entry, this is contradictory to the results in Fig. S4 and Fig 2. It is difficult to understand callose functioned in virus reproduction in 3 days post virus inoculation. And authors also avoided to explain this mechanism.

(2) Missing significant data. For example, the phenotypes of the transgenic plants, the RSV titers in the transgenic plants (OsTLP OE, ostlp). The staining of callose deposition were also hard to convince. The evidence about RSV NP-LssaCA-OsTLP tripartite interaction to enhance OsTLP enzymatic activity is not enough.

(3) Figure 4a, there was the LssaCA signal in the fourth lane of pull-down data. Did MBP also bind LssaCA? The characterization of pull-down methods was rough a little bit. The method of GST pull-down and MBP pull-down should be characterized more in more detail.

<https://doi.org/10.7554/eLife.88132.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

In this study, the authors identify an insect salivary protein participating viral initiate infection in plant host. They found a salivary LssaCA promoting RSV infection by interacting with OsTLP that could degrade callose in plants. Furthermore, RSV NP bond to LssaCA in salivary glands to form a complex, which then bond to OsTLP to promote degradation of callose.

The story focus on tripartite virus-insect vector-plant interaction and is interesting. However, the study is too simple and poor-conducted. The conclusion is also overstated due to unsolid findings.

We thank the reviewer for their constructive feedback. We have conducted additional experiments to strengthen our results and conclusions as detailed below:

(1) The comparison between vector inoculation and microinjection involves multiple confounding factors that could affect the experimental results, including salivary components, RSV inoculation titers, and the precision of viral deposition. The differential outcomes could be attributed to these various factors rather than definitively demonstrating the necessity of salivary factors. Therefore, we have removed this comparison from the revised manuscript and instead focused on elucidating the specific mechanisms by which LssaCA facilitates viral infection.

(2) We conducted new experiments to assess the function of LssaCA enzymatic activity in mediating RSV infection. Additional experiments revealed that OsTLP enzymatic activity is highly pH-dependent, with increased activity as pH decreases from 7.5 to 5.0 (Fig. 3H). However, the LssaCA-OsTLP interaction at pH 7.4 significantly enhanced OsTLP enzymatic activity without requiring pH changes. These results demonstrate that LssaCA-OsTLP protein interactions are crucial for mediating RSV infection. In contrast to pH-dependent mechanisms, our study demonstrated that LssaCA's biological function in mediating RSV infection is at least partially, if not completely, independent of its enzymatic activity. We have added these new results into the revised manuscript (Lines 220-227). We have also added a comprehensive discussion comparing the aphid CA mechanism described by Guo et al. (2023 doi.org/10.1073/pnas.2222040120) with our findings in the revised manuscript (Lines 350-371).

(3) We have repeated majority of callose deposition experiments, providing clearer images (Figures 5-6). In addition to aniline blue staining, we quantified callose concentrations using a plant callose ELISA kit to provide more precise measurements (Figure 5A, I, 6A, C and S8A). We utilized RT-qPCR to measure callose synthase expression in both feeding and non-feeding

areas, confirming that callose synthesis was induced specifically in feeding regions, leading to localized callose deposition (Figures 5D-G and S8B-E). For sieve plate visualization, we examined longitudinal sections, which revealed callose deposition in sieve plates during SBPH feeding and RSV infection (Figure S7).

(4) We generated OsTLP mutant rice seedlings (*ostlp*) and use this mutant to directly demonstrate that LssaCA mediates callose degradation in planta through enhancement of OsTLP enzymatic activity (Lines 288-302 and Figure 6).

(5) We produced LssaCA recombinant proteins in sf9 cells to ensure full enzymatic activity and constructed a comprehensive CA mutant protein, in which all seven residues constituting the enzymatic active center mutated (LssaCA^{H111D}, LssaCA^{N139H}, LssaCA^{H141D}, LssaCA^{H143D}, LssaCA^{E153H}, LssaCA^{H166D}, LssaCA^{T253E}) (Fig. S1B). This LssaCA mutant protein demonstrated complete loss of enzymatic activity (Fig. 1C).

Major comments:

(1) The key problem is that how long the LssCA functioned for in rice plant. Author declared that LssCA had no effect on viral initial infection, but on infection after viral inoculation. It is unreasonable to conclude that LssCA promoted viral infection based on the data that insect inoculated plant just for 2 days, but viral titer could be increased at 14 days post-feeding. How could saliva proteins, which reached phloem 12-14 days before, induce enough TLP to degrade callose to promote virus infection? It was unbelievable.

We appreciate your insightful comment and acknowledge that our initial description may have been unclear. We agree that salivary proteins would not present in plant tissues for two weeks post-feeding or post-injection. Our intention was to clarify that when salivary proteins enhance RSV infection, this initial enhancement leads to sustained high viral loads. We measured viral burden at 14 days post-feeding or post-injection because this is the common measurement time point when viral titers are sufficiently high for reliable detection by qRT-PCR or western blotting. We have clarified this rationale in the revised manuscript (Lines 155-157).

To determine the actual persistence of LssaCA in plant tissues, we conducted additional experiments where insects were allowed to feed on a defined area of rice seedlings for two days. We then monitored LssaCA protein levels at 1 and 3 days after removing the insects. Western blotting analysis revealed that LssaCA protein levels decreased post-feeding and remained detectable at 3 days post-feeding. These results are presented in Figure 2H and described in detail in Lines 184-193.

(2) Lines 110-116 and Fig. 1, the results of viruliferous insect feeding and microinjection with purified virus could not conclude the saliva factor necessary of RSV infection, because these two tests are not in parallel and comparable. Microinjection with salivary proteins combined with purified virus is comparable with microinjection with purified virus.

We thank the reviewer's insightful comment. We agree that "the results of viruliferous insect feeding and microinjection with the purified virus could not conclude the saliva factor necessary of RSV infection". However, due to the technical difficulty in collecting sufficient quantities of salivary proteins to conduct the microinjection experiment, we have removed these results from the revised manuscript.

(3) The second problem is how many days post viruliferous insect feeding and microinjection with purified virus did author detect viral titers? in Method section,

authors declared that viral titers was detected at 7-14 days post microinjection. Please demonstrate the days exactly.

We thank the reviewer's insightful comment. We typically measured RSV infection levels at both 7- and 14-days post-microinjection. However, since the midrib microinjection experiments have been removed from the revised manuscript, this methodology has also been removed accordingly.

(4) The last problem is that how author made sure that the viral titers in salivary glands of insects between two experiments was equal, causing different phenotype of rice plant. If not, different viral titers in salivary glands of insects between two experiments of course caused different phenotype of rice plant.

We thank the reviewer's comment. When we compared the effects of LssaCA deficiency on RSV infection of rice plants, we have compared the viral titers in the insect saliva and salivary glands. The results indicated that the virus titers in both tissues have not changed by LssaCA deficiency, suggesting that the viruses inoculated into rice phloem by insects of different treatments were comparable. Please refer to the revised manuscript Figures 2D-G and Lines 161-173.

(5) The callose deposition in phloem can be induced by insect feeding. In Fig. 5H, why was the callose deposition increased in the whole vascular bundle, but not phloem? Could the transgenic rice plant directional express protein in the phloem? In Fig. 5, why was callose deposition detected at 24 h after insect feeding? In Fig. 6A, why was callose deposition decreased in the phloem, but not all the cells of the of TLP OE plant? Also in Fig.6A and B, expression of callose synthase genes was required.

We thank the reviewer for these insightful comments.

(1) Figure 5. The callose deposition increased in multiple cells within the vascular bundle, including sieve tubes, parenchymatic cells, and companion cells. While callose deposition was detected in other parts of the vascular bundle, no significant differences were observed between treatments in these regions, indicating that in response to RSV infection and other treatments, altered callose deposition mainly occurred in phloem cells. Please refer to the revised 5B, 5J, 6B, and 6D.

(2) Transgenic plant expression. The OsTLP-overexpressing transgenic rice plants express TLP proteins in various cells under the control of CaMV 35S promoter, rather than being directionally expressed in the phloem. However, since TLP proteins are secreted, they are potentially transported and concentrated in the phloem where they can degrade callose.

(3) Figure 5. The 24-hour time point for callose deposition detection was selected based on established protocols from previous studies. According to Hao et al. (Plant Physiology 2008), callose deposition increased during the first 3 days of planthopper infestation and decreased after 4 days. Additionally, Ellinger and Voigt (Ann Bot 2014) demonstrated that callose visualization typically begins 18-24 hours after treatment, making 24 hours an optimal detection time point.

(4) Figure 6, Phloem-specific changes. Similar to Figure 5, while callose deposition was detected in other parts of vascular bundle, significant differences between treatments were mainly observed in phloem cells, indicating that RSV infection specifically affects callose deposition in phloem tissue.

(5) Callose synthase gene expression. We performed RT-qPCR analysis to measure the expression levels of callose synthase genes. The results indicated that OsTLP overexpression

did not significantly alter the mRNA levels of these genes, regardless of RSV infection status in SBPH.

Reviewer #2 (Public Review):

*There is increasing evidence that viruses manipulate vectors and hosts to facilitate transmission. For arthropods, saliva plays an essential role for successful feeding on a host and consequently for arthropod-borne viruses that are transmitted during arthropod feeding on new hosts. This is so because saliva constitutes the interaction interface between arthropod and host and contains many enzymes and effectors that allow feeding on a compatible host by neutralizing host defenses. Therefore, it is not surprising that viruses change saliva composition or use saliva proteins to provoke altered vector-host interactions that are favorable for virus transmission. However, detailed mechanistic analyses are scarce. Here, Zhao and coworkers study transmission of rice stripe virus (RSV) by the planthopper *Laodelphax striatellus*. RSV infects plants as well as the vector, accumulates in salivary glands and is injected together with saliva into a new host during vector feeding.*

The authors present evidence that a saliva-contained enzyme - carbonic anhydrase (CA) - might facilitate virus infection of rice by interfering with callose deposition, a plant defense response. In vitro pull-down experiments, yeast two hybrid assay and binding affinity assays show convincingly interaction between CA and a plant thaumatin-like protein (TLP) that degrades callose. Similar experiments show that CA and TLP interact with the RSV nuclear capsid protein NT to form a complex. Formation of the CA-TLP complex increases TLP activity by roughly 30% and integration of NT increases TLP activity further. This correlates with lower callose content in RSV-infected plants and higher virus titer. Further, silencing CA in vectors decreases virus titers in infected plants.

(1) Interestingly, aphid CA was found to play a role in plant infection with two non-persistent non-circulative viruses, turnip mosaic virus and cucumber mosaic virus (Guo et al. 2023 doi.org/10.1073/pnas.2222040120), but the proposed mode of action is entirely different.

We appreciate the reviewer's insightful comment and have carefully examined the cited publication. The study by Guo et al. (2023) elucidates a distinct mechanism for aphid-mediated transmission of non-persistent, non-circulative viruses (turnip mosaic virus and cucumber mosaic virus). In their model, aphid-secreted CA-II in the plant cell apoplast leads to H⁺ accumulation and localized acidification. This trigger enhanced vesicle trafficking as a plant defense response, inadvertently facilitating virus translocation from the endomembrane system to the apoplast.

In contrast to these pH-dependent mechanisms, our study demonstrated that LssaCA's biological function in mediating RSV infection is, if not completely, at least partially independent of its enzymatic activity. We performed additional experiments to reveal that OsTLP enzymatic activity is highly pH-dependent and exhibits increased enzymatic activity as pH decreases from 7.5 to 5.0 (Fig. 3H); however, the LssaCA-OsTLP interaction occurring at pH 7.4 significantly enhanced OsTLP enzymatic activity without any change in buffer pH (Fig. 3G). These results demonstrate the crucial importance of LssaCA-OsTLP protein interactions, rather than enzymatic activity alone, in mediating RSV infection.

We have incorporated these new experimental results and added a comprehensive discussion comparing the aphid CA mechanism described by Guo et al. (2023) with our findings in the revised manuscript. Please refer to Figures 3G-H, Lines 220-227 and 350-371 for detailed information.

(2) While this is an interesting work, there are, in my opinion, some weak points. The microinjection experiments result in much lower virus accumulation in rice than infection by vector inoculation, so their interpretation is difficult.

We acknowledge the reviewer's concern regarding the lower virus accumulation observed in microinjection experiments compared to vector-mediated inoculation. We have removed these experiments from the revised manuscript. To address the core question raised by these experiments, we have conducted new experiments that directly demonstrate the importance of LssaCA-OsTLP protein-protein interactions in mediating RSV infection. These results demonstrate the crucial importance of LssaCA-OsTLP protein interactions, rather than enzymatic activity alone, in mediating RSV infection. Additionally, we have incorporated a comprehensive discussion examining carbonic anhydrase activity, pH homeostasis, and viral infection. Please refer to the detailed experimental results and discussion in the sections mentioned in our previous response (Figures 3G-H, Lines 220-227 and 350-371).

(3) Also, the effect of injected recombinant CA protein might fade over time because of degradation or dilution.

We appreciate the reviewer's insightful comment. This is indeed a valid concern that could affect the interpretation of microinjection results. To address the temporal dynamics of CA protein presence in planta, we conducted time-course experiments to monitor the retention of naturally SBPH-secreted CA proteins in rice plants. Our analysis at 1- and 3- days post-feeding (dpf) revealed that CA protein levels decreased progressively following SBPH feeding, but could also be detected at 3dpf (Fig. 2H). Please refer to Figures 2H and lines 184-193 for detailed information.

(4) The authors claim that enzymatic activity of CA is not required for its proviral activity. However, this is difficult to assess because all CA mutants used for the corresponding experiments possess residual activity.

We appreciate the reviewer's insightful comment. We constructed a comprehensive CA mutant protein in which all seven residues constituting the enzymatic active center mutated (LssaCA^{H111D}, LssaCA^{N139H}, LssaCA^{H141D}, LssaCA^{H143D}, LssaCA^{E153H}, LssaCA^{H166D}, LssaCA^{T253E}) (Fig. S1B). This LssaCA mutant protein demonstrated complete loss of enzymatic activity (Fig. 1C). However, since we have removed the recombinant CA protein microinjection experiments from the revised manuscript, we lack sufficient direct evidence to definitively demonstrate that CA enzymatic activity is dispensable for its proviral function. To address the core question raised by these experiments, we have conducted new experiments that provide direct evidence for the importance of LssaCA-OsTLP protein-protein interactions in mediating RSV infection. Additionally, we have incorporated a comprehensive discussion examining carbonic anhydrase activity, pH homeostasis, and viral infection. Please refer to the detailed experimental results and discussion in the sections mentioned in our previous response (Figures 3G-H, Lines 220-227 and 350-371).

(5) It remains also unclear whether viral infection deregulates CA expression in planthoppers and TLP expression in plants. However, increased CA and TLP levels could alone contribute to reduced callose deposition.

We have compared LssaCA mRNA levels in RSV-free and RSV-infected *L. striatellus* salivary glands, which indicated that RSV infection does not significantly affect LssaCA expression (Figure 1J). By using RSV-free and RSV-infected *L. striatellus* to feed on rice seedlings, we clarified that RSV infection does not affect TLP expression in plants (Figure 5H).

Reviewer #1: (Recommendations For The Authors):

Other comments:

(1) Most data proving viral infection and LssaCA expression were derived from qPCR assays. Western blot data are strongly required to prove the change at the protein level.

We agree that western blot data are required to prove the change at the protein level. In the revised manuscript, we have added western-blotting results (Figures 1F, 1I, 2C, 2J, and S6).

(2) Line 145, data that LssaCA was significantly downregulated should be shown.

Thank you and the data has been added to the revised manuscript. Please refer to Line 165 and Figure 2D.

(3) Lines 159-161, how did authors assure that the dose of recombinant LssCA was closed to the release level of insect feeding, but not was excessive? How did author exclude the possibility of upregulated RSV titer caused by excessive recombinant LssCA?

We appreciate this important concern regarding dosage controls. While microinjection of recombinant proteins typically yields viral infection levels significantly lower than those achieved through natural insect feeding, higher protein concentrations are often required to achieve high viral infection levels. In this experiment, we compared RSV infection levels following microinjection of BSA+RSV versus LssaCA+RSV, with the expectation that any observed upregulation in RSV titer would be specifically attributable to recombinant LssaCA rather than excessive protein dosing. However, given the low RSV infection levels observed with viral microinjection, we have removed their corresponding results from the revised manuscript.

(4) Lines 124-125, recombinantly expressed LssaCA protein should be underlined, but not the LssaCA protein itself.

We have clearly distinguished recombinantly expressed LssaCA from endogenous LssaCA protein throughout the manuscript, ensuring that all references to recombinant proteins are properly labeled as such.

(5) LssaCA expression in salivary glands of viruliferous and nonviruliferous insects is required. LssaCA accumulation in rice plant exposed to viruliferous and nonviruliferous insects is also required.

We have measured LssaCA mRNA levels in salivary glands of viruliferous and nonviruliferous insects (Figure 1J), and protein levels in rice plant exposed to viruliferous and nonviruliferous insects (Figure 1I).

(6) Fig. 4G, the enzymatic activities of OsTLP were too low compared with that in Fig. 4E and Fig. 7E. Why did the enzymatic activities of the same protein show so obvious difference?

We apologize for the error in Fig. 4G. The original data presented relative fold changes between OsTLP+BSA and OsTLP+LssaCA treatment, with OsTLP+BSA normalized to 1.0 and OsTLP+LssaCA values expressed as fold changes relative to this baseline. However, the Y-axis was incorrectly labeled as “ β -1,3-glucanase (units mg^{-1})”, which suggested absolute enzymatic activity values. We have now corrected the figure (revised Figure 3G) to display the actual

absolute enzymatic activity values with the appropriate Y-axis label “ β -1,3-glucanase (units mg^{-1})”.

(7) Fig. 7E, was the LssaCA + NP and LssaCA + GST quantified?

Yes, all proteins were quantified, and enzymatic activity values were calculated and expressed as units per milligram of proteins (units mg^{-1}).

Minor comments:

(1) The keywords: In fact, the LssaCA functioned during initial viral infection in plant, but not viral horizontal transmission.

We appreciate the reviewer’s insightful comment. We have revised the manuscript title to “Rice stripe virus utilizes an Laodelphax striatellus salivary carbonic anhydrase to facilitate plant infection by direct molecular interaction” and changed the keyword from “viral horizontal transmission” to “viral infection of plant”.

(2) Fig. 2A, how about testes? Was this data derived from female insects? Fig. 2C, is the saliva collected from nonviruliferous insects? Fig. 2E, what is the control?

We appreciate the reviewer’s insightful comments.

(1) Fig. 2A: The data present mean and SD calculated from three independent experiments, with 5 tissue samples per experiment. Since 3rd instar nymphs were used for feeding experiments in this study, we also used 3rd instar RSV-free nymphs to measure gene expression in guts, salivary glands and fat bodies. R-body represents the remaining body after removing these tissues. Female insects were used to measure gene expression in ovaries, and gene expression in testes was also added. We have added this necessary information to the revised manuscript (please refer to new Figure 1F and Lines 402-403).

(2) Fig. 2C: Yes, saliva was collected from nonviruliferous insects.

(3) Fig. 2E: The control consisted of 100 mM PBS, as described in the experimental section (Lines 643-644): “A blank control consisted of 2 mL of 100 mM PBS (pH 7.0) mixed with 1 mL of 3 mM p-NPA.” In the revised manuscript, we recombinantly expressed LssaCA and its mutant proteins in both sf9 cells and E.coli. Therefore, we have used the mutant proteins as controls to demonstrate specific enzymatic activity. Please refer to Figure 1C, Lines 115-122 and 621-635 for detailed information.

(3) Some figure labeling appeared unprofessional. For example, “a-RSV”, “loading” in Fig. 1, “W-saliva”, “G-saliva” in Fig. 2, and so on, the related explanations were absent.

We appreciate the reviewer’s insightful comments. We have thoroughly reviewed all figures to ensure professional labels. Specifically, we have:

(1) Used proper protein names to label western blots and clearly explained the antibodies used for protein detection.

(2) Provided comprehensive explanations for all abbreviations used in figures within the corresponding figure legends.

(3) Ensured consistent and clear labeling throughout all figures.

Please refer to the revised Figures 1-3 for these corrections.

(4) Lines 83-84, please cite references on callose preventing viral movement. I do not think the present references were relevant.

We have added a more relevant reference (Yue et al., 2022, Line 82), which revealed that palmitoylated yb promotes virus cell-to-cell movement by interacting with NbREM1 to inhibit callose deposition at plasmodesmata.

(5) The background of transgenic plants of OsTLP OE should be characterized. And the overexpression of OsTLP should be shown. Which generation of OsTLP OE did authors use?

The background of transgenic plants of OsTLP OE and its generation used have been shown in the “Materials and methods” section (Line 782-786) and has been mentioned in the main text (Line 214). T² lines have been selected for further analysis (Line 789).

(6) Fig. 5A, the blank, which derived from plants without exposure to insect, was absent.

We appreciate the reviewer’s insightful comments. We have added the non- fed control in the revised Figure 5A-C.

(7) Fig. 7A, the nonviruliferous insects were required to serve as a control.

Immunofluorescence localization of RSV and LssaCA in uninfected *L. striatellus* salivary glands have been added to the revised manuscript (Figure S2).

(8) The manuscript needs English language edit.

The manuscript has undergone comprehensive English language editing to improve clarity, grammar, and overall readability.

Reviewer #2 (Recommendations For The Authors):

(1) The first experiment compares vector inoculation vs microinjection of RSV in tissue. I am not sure that your claim (saliva factors are necessary for inoculation) holds, because the vector injects RSV directly into the phloem, whereas microinjection is less precise and you cannot control where exactly the virus is deposited. However, virus deposited in other tissues than the phloem might not replicate, and indeed you observe, compared to natural vector inoculation, highly reduced virus titers.

We appreciate the reviewer’s insightful comments. We agree that the comparison between vector inoculation and microinjection involves multiple confounding factors that could affect the experimental results, including salivary components, RSV inoculation titers, and the precision of viral deposition. As the reviewer correctly points out, the differential outcomes could be attributed to these various factors rather than definitively demonstrating the necessity of salivary factors. Therefore, we have removed this comparison from the revised manuscript and instead focused on elucidating the specific mechanisms by which LssaCA facilitates viral infection.

(2) Next the authors show that a carbonic anhydrase (CA) that they previously detected in saliva is functional and secreted into rice. I assume this is done with non-infected insects, but I did not find the information. Silencing the CA reduces virus titers in inoculated plants at 14 dpi, but not in infected planthoppers. At 1 dpi, there is no difference in RSV titer in plants inoculated with CA silenced planthoppers or control hoppers. To see a

direct effect of CA in virus infection, purified virus is injected together with a control protein or recombinant CA into plants. At 14 dpi, there is about double as much virus in the CA-injected plants, but compared to authentic SBPH inoculation, titers are 20,000 times lower. Actually, I believe it is not very likely that the recombinant CA is active or present so long after initial injection.

We appreciate the reviewer's insightful comments.

(1) Our previous study identified the CA proteins from RSV-free insects. We have added this information to the revised manuscript (Line 110).

(2) We acknowledge the reviewer's concern regarding the lower virus accumulation observed in microinjection experiments compared to vector-mediated inoculation. We have removed these experiments from the revised manuscript and instead focused on elucidating the specific mechanisms by which LssaCA facilitates viral infection.

(3) We didn't intend to suggest that LssaCA proteins presented for 14 days post-injection. We measured viral titers at 14 days post-feeding or post-injection because this is the common measurement time point when viral titers are sufficiently high for reliable detection by RT-qPCR or western blotting. We have clarified this rationale in the revised manuscript (Lines 155-157). To determine the actual persistence of LssaCA in plant tissues, we monitored LssaCA protein levels at 1 and 3 dpf. Western blotting analysis revealed that LssaCA protein levels decreased post-feeding and remained detectable at 3 dpf. These results are presented in Figure 2H and described in detail in Lines 184-193.

(3) Then the authors want to know whether CA activity is required for its proviral action and single amino acid mutants covering the putative active CA site are created. The recombinant mutant proteins have 30-70 % reduced activity, but none of them has zero activity. When microinjected together with RSV into plants, RSV replication is similar as injection with wild type CA. Since no knock-out mutant with zero activity is used, it is difficult to judge whether CA activity is unimportant for viral replication, as claim the authors.

We appreciate the reviewer's insightful comment. We constructed a comprehensive CA mutant protein in which all seven residues constituting the enzymatic active center mutated (LssaCA^{H111D}, LssaCA^{N139H}, LssaCA^{H141D}, LssaCA^{H143D}, LssaCA^{E153H}, LssaCA^{H166D}, LssaCA^{T253E}) (Fig. S1B). This LssaCA mutant protein demonstrated complete loss of enzymatic activity (Fig. 1C). However, since we have removed the recombinant CA proteins microinjection experiments from the revised manuscript, we lack sufficient direct evidence to definitively demonstrate that CA enzymatic activity is dispensable for its proviral function. To address the core question raised by these experiments, we have conducted new experiments that provide direct evidence for the importance of LssaCA-OsTLP protein-protein interactions in mediating RSV infection. Additionally, we have incorporated a comprehensive discussion examining carbonic anhydrase activity, pH homeostasis, and viral infection. Please refer to the detailed experimental results and discussion in the sections mentioned in our previous response (Figures 3G-H, Lines 220-227 and 350-371).

(4) Next a yeast two hybrid assay reveals interaction with a thaumatin-like rice protein (TLP). It would be nice to know whether you detected other interacting proteins as well. The interaction is confirmed by pulldown and binding affinity assay using recombinant proteins. The kD is in favor of a rather weak interaction between the two proteins.

We have added a list of rice proteins that potentially interact with LssaCA (Table S1) and have measured interactions with additional proteins (unpublished data). Despite the relatively

weak binding affinity, the functional significance of the LssaCA-OsTLP interaction in enhancing TLP enzymatic activity is substantial.

(5) Then the glucanase activity of TLP is measured using recombinant TLP-MBP or in vivo expressed TLP. It is not clear to me which TLP is used in Fig. 4G (plant-expressed or bacteria-expressed). If it is plant-expressed TLP, why is its basic activity 10 times lower than in Fig. 4F?

Fig. 4G is the Fig. 3G in the revised manuscript. A *E. coli*-expressed TLP protein has been used. We apologize for the error in our original Fig. 4G. The original data presented relative fold changes between OsTLP+BSA and OsTLP+LssaCA treatment, with OsTLP+BSA normalized to 1.0 and OsTLP+LssaCA values expressed as fold changes relative to this baseline. However, the Y-axis was incorrectly labeled as “ β -1,3-glucanase (units mg^{-1})”, which suggested absolute enzymatic activity values. We have now corrected the figure to display the actual absolute enzymatic activity values with the appropriate Y-axis label “ β -1,3-glucanase (units mg^{-1})”.

(6) There is also a discrepancy in the construction of the transgenic rice plants: did you use TLP without signal peptide or full length TLP? If you used TLP without signal peptide, you should explain why, because the wild type TLP contains a signal peptide.

We cloned the full-length OsTLP gene including the signal peptide sequence (Line 782 in the revised manuscript).

(7) The authors find that CA increases glucanase activity of TLP. Next the authors test callose deposition by aniline blue staining. Feeding activity of RSV-infected planthoppers induces more callose deposition than does feeding by uninfected insects. In the image (Fig. 5A) I see blue stain all over the cell walls of xylem and phloem cells. Is this what the authors expect? I would have expected rather a patchy pattern of callose deposition on cell walls. Concerning sieve plates, I cannot discern any in the image; they are easier to visualize in longitudinal sections than in transversal section as presented here.

We appreciate the reviewer’s insightful comment.

(1) Callose deposition pattern: While callose deposition was detected in other parts of the vascular bundle, significant differences between treatments were mainly observed in phloem cells, indicating that phloem-specific callose deposition is the primary response to RSV infection and SBPH feeding (Figures 5B and 5J).

(2) Sieve plate visualization: We have examined longitudinal sections to visualize sieve plates, which revealed callose deposition in sieve plates during SBPH feeding and RSV infection (Figure S7).

(3) Quantitative analysis: In addition to aniline blue staining, we quantified callose concentrations using a plant callose ELISA kit to provide more precise measurements (Figure 5A, 5I and S8A).

(4) Gene expression analysis: We utilized RT-qPCR to measure callose synthase expression in both feeding and non-feeding areas, confirming that callose synthesis was induced specifically in feeding regions, leading to localized callose deposition (Figures 5D-H).

These experimental results collectively demonstrate that RSV infection induces enhanced callose synthesis and deposition, with this response occurring primarily in phloem cells, including sieve plates, within feeding sites and their immediate vicinity.

(8) I do not quite understand how you quantified callose deposition (arbitrary areas?) with ImageJ. Please indicate in detail the analysis method.

We have added more detailed information for the methods to quantify callose deposition (Lines 673-678).

(9) More callose content is also observed by a callose ELISA assay of tissue extracts and supported by increased expression of glucanase synthase genes. Did you look whether expression of TLP is changed by feeding activity and RSV infection? Silencing CA in planthoppers increases callose deposition, which is inline with the observation that CA increases TLP activity.

We measured OsTLP expression following feeding by RSV-free or RSV-infected SBPH and found that gene expression was not significantly affected by either insect feeding or RSV infection. These results have been added to the revised manuscript (Lines 275-277 and Figure 5H).

(10) Next, callose is measured after feeding of RSV-infected insects on wild type or TLP-overexpressing rice. Less callose deposition (after 2 days) and more virus (after 14 days) is observed in TLP overexpressors. I am missing a control in this experiment, that is feeding of uninfected insects on wild type or TLP overexpressing rice, where I would expect intermediate callose levels.

We appreciate the reviewer's insightful comment and fully agree with the prediction. In the revised manuscript, we have constructed ostlp mutant plants and conducted additional experiments to further clarify how callose deposition is regulated by insect feeding, RSV infection, LssaCA levels, and OsTLP expression. Specifically:

(1) Both SBPH feeding and RSV infection induce callose deposition, with RSV-infected insect feeding resulting in significantly higher callose levels compared to RSV-free insect feeding (Fig. 5A-C).

(2) LssaCA enhances OsTLP enzymatic activity, thereby promoting callose degradation (Fig. 5I-K).

(3) OsTLP-overexpressing (OE) plants exhibit lower callose levels than wild-type (WT) plants, while ostlp mutant plants show higher callose levels than WT (Fig. 6A-B).

(4) In ostlp knockout plants, LssaCA no longer affects callose levels, indicating that OsTLP is required for LssaCA-mediated regulation of callose (Fig. 6C-D).

These additional data address the reviewer's concern and support the conclusion that OsTLP plays a central role in modulating callose levels in response to RSV infection and insect feeding.

(11) Next the authors test for interaction between virions and CA. Immunofluorescence shows that RSV and CA colocalize in salivary glands; in my opinion, there is partial and not complete colocalization (Fig. 7A).

We agree with the reviewer's observation. CA is primarily produced in the small lobules of the principal salivary glands, while RSV infects nearly all parts of the salivary glands. In regions where RSV and CA colocalize within the principal glands, the CA signal appears sharper than that of RSV, likely due to the relatively higher abundance of CA compared to RSV in these areas. This may explain the partial, rather than complete, colocalization observed in our original Figure 7A. In the revised manuscript, please refer to Figure 1A.

(12) Pulldown experiments with recombinant RSV NP capsid protein and CA confirm interaction, binding affinity assays indicate rather weak interaction between CA and NP.

Likewise in pull-down experiments, interaction between NP, CA and TLP is shown. Finally, in vitro activity assays show that activity of preformed TLP-CA complexes can be increased by adding NP; activity of TLP alone is not shown.

We performed two independent experiments to confirm the influence on TLP enzymatic activity by LssaCA or by the LssaCA-RSV NP complex. In the first experiment, we compared the enhancement of TLP activity by LssaCA using TLP alone as a control (Figure 3G). In the second experiment examining the LssaCA-RSV NP complex effect on TLP activity, we used the LssaCA-TLP combination as the baseline control rather than TLP alone (Figure 4B), since we had already established the LssaCA enhancement effect in the previous experiment.

(13) For all microscopic acquisitions, you should indicate the exact acquisition conditions, especially excitation and emission filter settings, kind of camera used and objectives. Use of inadequate filters or of a black & white camera could for example be the reason why you observe a homogeneous cell wall label in the aniline blue staining assays. Counterstaining cell walls with propidium iodide might help distinguish between cell wall and callose label.

Thank you for your insightful suggestions. We have added the detailed information to the revised manuscript (Lines 656-659 and 673-678).

(14) You should provide information whether CA is deregulated in infected planthoppers, as this could also modify its mode of action.

We have compared LssaCA mRNA levels in RSV-free and RSV-infected *L. striatellus* salivary glands. The results indicated that RSV infection does not significantly affect LssaCA expression (Figure 1J).

(15) You should show purity of the proteins used for affinity binding measurements.

We have included SDS-PAGE results of purified proteins in the revised manuscript (Figure S3).

(16) L 39: Not all arboviruses are inoculated into the phloem.

Thank you. We have revised this description (Lines 40, 73, 95 and 97).

(17) L 76: Watery saliva is also injected in epidermis and mesophyll cells.

Thank you. We have revised this description (Line 73).

(18) L 79: What do you mean by "avirulent gene"?

Thank you for your valuable comments. We have revised this description as "certain salivary effectors may be recognized by plant resistance proteins to induce effector-triggered immunity". Please refer to Lines 76-77 for detail.

(19) L 128: Please add delivery method.

Thank you. We have added the delivery methods (Line 134).

(20) L 195: Please explain "MST".

Explained (Line 124). Thank you.

| (21) L 203: *Please add the plant species overexpressing TLP.*

Added (Line 214). Thank you.

| (22) L 213: *Callose deposition has also a role against phloem-feeding insects.*

We appreciate the reviewer's insight comment. We have added this information to the revised manuscript (Line 252).

| (23) L 626: *What is a "mutein"?*

"mutein" is an abbreviation for mutant proteins. Since the recombinant protein microinjection experiments have been removed from the revised manuscript, the term "mutein" has also been removed. For all other instances, we now use the full term "mutant proteins".

| (24) Fig. 1E: *what is "loading"? You should rather show here and elsewhere (or add to supplement) complete protein gels and Western blot membranes and not only bands of interest.*

Thank you for your valuable suggestion. Although Figure 1E has been removed from the revised manuscript, we have carefully reviewed all figures to ensure that the term "loading" has been replaced with the specific protein names where appropriate.

| (25) Fig. 2C: *Please indicate which is the blot and which is the silver stained gel and add mass markers in kDa to the silver stained gel.*

Thank you for your suggestion. We have revised figure to include labeled silver-stained gels with indicated molecular weight markers (Figure 1H in the revised manuscript).

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